Conflated Random Slopes in Multilevel Analysis

by

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A thesis submitted in partial fulfillment
of the requirements for the Master of Arts
degree in Psychological and Quantitative Foundations in the
Graduate College of
The University of Iowa

December 2025

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ACKNOWLEDGMENTS

This work would not have been possible without the support and work of Lesa Hoffman, who has contributed to my academic, professional, and pedagogical development. Thank you, Lesa, for being the mentor I needed for this long journey. Thank you to Professor Jonathan Templin for his support and teachings, both in the classroom and in our personal conversations. Thank you, Jonathan, for helping me navigate complex topics without too much fear. Thank you to Professor Ariel Aloe for the time and effort he dedicated to my academic and personal development. Thank you, Ariel, for nurturing my critical thinking and for your sincerity. Thank you to Professor Carlos Rodriguez, who has contributed to my professional and academic career for years. Thank you, Carlos, for your selfless work, your support, and your friendship.

I would like to acknowledge the company and support that my close friends provided me on many occasions, which was important in alleviating my mental health concerns. Finally, I would like to acknowledge the unconditional support of Claudia, who has affectionately helped me achieve the emotional and affective stability I have needed to work on this and other projects.

ABSTRACT

Multilevel analysis is widely used for hierarchical data (e.g., individuals within clusters), allowing researchers to examine how individual- and cluster-level (i.e., level 1 and level 2) attributes relate to individual outcomes. A well-known issue concerns conflated fixed slopes, which arise when a level 1 predictor with systematic variance across clusters is modeled with a single fixed effect at the individual level, conflating its within- and between-cluster slopes. Less recognized, however, is the problem of conflated random slopes that occurs when the sources of heterogeneity introduced by the within- and between-cluster components of a level 1 predictor are not separated (i.e., the fixed effects of the predictor are separated across levels, but not their random effects). A conflated random slope is a blend of slope heterogeneity and intercept heteroscedasticity, producing misleading conclusions about between-cluster differences. Proper parameterization of the random slopes is necessary to avoid this issue, either using cluster-mean-centering (CMC) or constant-centering (CC). Despite its importance, the practical consequences of conflated random slopes remain rarely discussed in psychology and educational research. This study addresses that gap using a Monte Carlo simulation comparing three models that estimate unconflated random slopes and one widely used model that estimates conflated random slopes. The models differed in their parameterization of the random slope for a level 1 predictor: The within-only model had a random slope for the CMC level 1 predictor only; the within–between model had random slopes for both the CMC level 1 predictor and the CC level 2 predictor; the within–contextual model had random slopes for the CC level 1 predictor and the CC level 2 predictor; and the random-conflated model had a random slope for the CC level 1 predictor only. The simulation factors were level 1 sample size (5 and 30), level 2 sample size (25 and 50), random slope reliability (0.20 and 0.60), and the ratio of intercept heteroscedasticity to slope

heterogeneity (0.5, 1.0, 1.5). After examining model convergence, model performance was assessed in terms of bias in fixed and random effects, power to detect random slopes, and accuracy of standard errors. Results showed clear trade-offs. The random-conflated model had very high convergence (> .97) even in small samples, but it consistently underestimated level 1 slope variance when within- and between-cluster variances differed, yielding downwardly biased variance estimates. The within-only model correctly specified the random slope but imposed homogeneity in intercept variance across cluster means, leading to slightly biased SEs for level 2 fixed effects. By contrast, the within–between and within–contextual models—theoretically correct specifications—had substantially lower convergence rates (.38–.73) in small samples but generally outperformed the simpler alternatives in parameter estimation and SE accuracy in larger sample sizes. Regarding power, all four models showed high capability to detect non-zero level 1 random slope variance. However, power to detect level 2 random slope variance in the within–between and within–contextual models was modest (.39 and .58, respectively), especially when sample sizes at level 1 and level 2 were small. This highlights the tension in practice between model complexity and the available information for the model from the data. Overall, our results show that estimating conflated random slopes distorts both fixed and random effect estimates, threatening valid inferences about between-cluster differences. When sample sizes permit, researchers may specify the within–between model, particularly if intercept heteroscedasticity matters. When data are limited or intercept heteroscedasticity is not of interest, the within-only model is a reasonable alternative to avoid random conflation, though with the trade-off of slightly biased SEs for level 2 fixed effects.

PUBLIC ABSTRACT

Multilevel analysis is a common statistical approach used to analyze hierarchically structured data (e.g., individuals in clusters), in which both individual (i.e., level 1) and cluster (i.e., level 2) attributes influence individual outcomes. In multilevel models, individual-level predictors often vary systematically between groups, and failing to separate their effects can lead to biased results. This study focuses on *conflated random slopes*, which are a blend of different sources of variability between clusters that distort cluster-to-cluster differences. To investigate this issue, a computer simulation was conducted to test four different modeling strategies under various data conditions. Findings showed that models with conflated random slopes often appeared to fit well but produced biased estimates of variability between clusters and misleading standard errors for fixed effects. Models that correctly separated within- and between-group differences provided more accurate results, though they required larger sample sizes to work reliably. The conclusions highlight a practical trade-off between model simplicity and statistical accuracy and offer guidance for researchers seeking to draw valid conclusions about variation across individuals and clusters.

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INTRODUCTION

Nested or clustered data are characterized by having multiple sampling dimensions, such students within classrooms or teachers within schools, which implies a hierarchical structure wherein the individual units of the same cluster tend to resemble each other more than do individual units from different clusters (Gelman & Hill, 2007). This dependence is a feature of the clustered data that cannot be omitted and that popular single-level linear regression models (simple and multiple; general and generalized) cannot easily accommodate in their structure given their requirement of independent residuals (Finch et al., 2019).

For clustered data, multilevel analysis has become a powerful method that translates each hierarchical sampling dimension in the outcome into a hierarchical level (e.g., level 1 and level 2 or individual-level and cluster-level) within a statistical model (McNeish & Wentzel, 2016). In these multilevel models, both fixed and random effects are used to capture the relationships of the predictors with the outcome considering intra- and inter-cluster differences. Conceptually, a multilevel model is useful for analyzing the contribution of individual and contextual attributes in the prediction of an outcome. Statistically, these models provide correct fixed effects, standard errors, and variance components that reflect the nested structure of the data (Hox et al., 2017).

Multilevel analysis is especially useful in the study of level 1 predictors in clustered data, which often exhibit systematic variation between clusters (i.e., differences in means across clusters) that should be removed from the raw predictor or controlled explicitly (Raudenbush & Bryk, 2002; Rights, 2022). For this reason, a common practice is to include the cluster means of the level 1 predictors at level 2 to estimate between-cluster or contextual fixed effects, depending on the interest of the researcher—who should have selected the corresponding centering technique for the level 1 predictor (McNeish, 2023; Yaremych et al., 2023). This disaggregates

the within- and between-cluster information in the raw level 1 predictor, which otherwise would have a *conflated* fixed effect in the model, i.e., an uninterpretable blend of its level-specific components (Preacher, 2011). This problem, widely known in the multilevel analysis literature (Raudenbush, 1989), is inherent to the parameterization of level 1 predictors and can lead to important estimation and interpretation issues (Hoffman & Walters, 2022).

It is highly recommended to disaggregate the within- and between-cluster information of level 1 predictors, as the fixed effects of both components may have different directions and magnitudes (Snijders & Bosker, 2012). The discussion on conflated fixed effects in multilevel analysis has been extensive; however, the corresponding discussion for the potential of conflated random effects has not been as notable. As elaborated below, conflated random effects are also linked to the parameterization of the level 1 predictors in the model, but this conflation results in an uninterpretable blend of slope heterogeneity and intercept heteroscedasticity across clusters (Rights & Sterba, 2023).

The purpose of this study is to examine the problem of conflated random effects in multilevel analysis and evaluate the models that can help overcome their potential biases. The organization of the remainder of this document is as follows. The first section reviews the background information necessary to understand the conflated random effects problem, its relationship with the level 1 predictors, and the bias it introduces in multilevel models. The second section describes the simulation design, models examined, and procedure. The third section highlights the main results and evidence to answer the research questions. Finally, the fourth section summarizes the implications of the findings, along with some limitations and future actions.

Theoretical Background on Multilevel Modeling

A multilevel model (MLM) is a useful method for solving some of the challenges that clustered data impose, especially when contextual attributes (i.e., cluster characteristics) are relevant to the study (Raudenbush & Bryk, 2002; Snijders & Bosker, 2012). Multilevel analysis, as developed from the combination of contextual analysis and the statistical theory of mixed models in the 1980s (De Leeuw et al., 2008), offers the advantage of modeling outcomes according to their hierarchical sampling dimensions. Using fixed and random effects, the relationships of both within- and between-cluster attributes with the outcome are estimated at each level at which they contain variability (Hox et al., 2017). Without loss of generality, in this study we will refer to individual outcomes with two sampling dimensions (and thus two levels, such as for individuals nested in clusters), but the issues discussed extend into more complex multilevel structures.

Fixed effects in an MLM capture constant relationships among predictors and the outcome, pooling information across level 1 and level 2 units in the data (Gelman & Hill, 2007). These coefficients are crucial for understanding the unique contribution of each predictor while accounting for both individual and cluster attributes (McNeish & Kelley, 2019). Fixed effects at level 1 describe the relationship of individual differences in the predictor to the outcome within clusters, whereas fixed effects at level 2 capture the relationship between cluster differences in the predictor and the outcome across clusters (Finch et al., 2019). Additionally, level 2 predictors derived from level 1 variables (i.e., cluster-mean predictors) can have contextual fixed effects, which represent the unique contribution of clusters above and beyond the fixed effects of their counterparts at level 1 (Yaremych et al., 2023).

Random effects in an MLM quantify the heterogeneity of fixed effects across clusters and help account for cluster dependence in the data by introducing correlation patterns in the variance–covariance matrix of units from the same cluster. Conceptually, random effects capture cluster-specific deviations from the fixed effects and can be used to predict cluster-specific outcomes (Hamaker & Muthén, 2020).

The Importance of Centering Level 1 Predictors

In multilevel analysis, level 1 predictors require special attention, given that their relationship with the outcome can differ across levels (Rights et al., 2020; Snijders, 2005). This is because level 1 predictors may also have information about clusters that must be disaggregated to avoid biases in the estimation and interpretation of their fixed effects (Curran et al., 2012; Preacher et al., 2010; Raudenbush & Bryk, 2002). Both components may have divergent information; not acknowledging such differences when specifying the model can lead to conceptual and statistical errors when estimating fixed and random effects (Hoffman & Walters, 2022).

As mentioned above, a common solution to this problem with respect to the fixed effects is to add the cluster means of the level 1 predictor to the model, so both predictors get separate fixed effects (Enders, 2013). Then, researchers center these variables at some meaningful constant to interpret the intercept, procedure known as constant-centering or CC (Hoffman, 2019). To illustrate this, let us consider an example scenario of students (level 1) nested in classrooms (level 2), as shown in Equation 1:

Level 1

$$y_{sc} = \beta_{0c} + \beta_{1c}(x_{sc}) + e_{sc}$$

$$e_{sc} \sim N(0, \sigma_e^2),$$

Level 2

$$\beta_{0c} = \gamma_{00} + \gamma_{01}^{c}(x_{.c}) + U_{0c}$$

$$\beta_{1c} = \gamma_{10}$$

$$U_{0c} \sim N\left(0, \tau_{U_0}^2\right), \tag{1}$$

in which the individual outcome y_{sc} (e.g., a test score) for the student s in classroom c is being predicted by x_{sc} at level 1 (e.g., student GPA centered at 3) and the corresponding cluster-mean predictor $x_{.c}$ at level 2 (i.e., classroom-mean GPA centered at 3). The β terms at level 1 are placeholders (or latent variables) composed of fixed and random effects at level 2. Specifically, β_{0c} is the predicted cluster-specific intercept composed of the fixed intercept γ_{00} (i.e., the conditional outcome mean across clusters when all predictors are zero), the contextual fixed effect γ_{01}^{c} for $x_{.c}$ (i.e., the unique contribution of the classroom GPA after controlling for the within-cluster fixed effect and values of x_{sc}), and U_{0c} (i.e., the cluster-specific deviation of the original cluster-mean outcome from γ_{00}). The term β_{1c} in Equation 1 is composed of the within-cluster fixed effect γ_{10} for x_{sc} (i.e., the expected change in y_{sc} per unit difference in student GPA within classrooms), which does not yet have a random slope, so it is constant across clusters.

Finally, the level-specific residuals e_{sc} and U_{0c} are both independent and normally distributed random variables with mean 0 and estimated variances σ_e^2 and $\tau_{U_0}^2$, respectively. Both level-specific residuals are not related to each other nor to the predictors at either level (Goldstein, 2011; Shaw & Flake, 2024). In addition, the variance components σ_e^2 and $\tau_{U_0}^2$ can be used to compute dependence and model effect sizes, such as the intraclass correlation (ICC =

$\tau^2_{U_0}/[\sigma^2_e + \tau^2_{U_0}]$) that reflects the proportion of total outcome variance attributable to between-cluster differences, as well as pseudo-R^2 measures that capture variance reduction attributable to level 1 or level 2 predictors across nested models.

Importantly, the model in Equation 1 has unconflated fixed effects, although the way in which it works does not directly separate the level-specific effects of the within- and between-cluster components. When using CC, both predictors have partial fixed effects that estimate the unique contribution of each one *controlling for the other* (Yaremych et al., 2023), which means that both predictors are necessary in the model to avoid conflated fixed slopes. Alternatively, researchers can use the centering-within-cluster (CWC) technique to parameterize the fixed effects of the level 1 predictors; CWC consists of centering the raw level 1 predictor using the corresponding cluster-mean predictor at level 2, which creates an orthogonal relationship across both variables and leads to estimate *purely* level-specific fixed effects (Enders & Tofighi, 2007). The main consequence is the interpretation of the new fixed effect at level 2, which is now a between-cluster fixed effect (i.e., the expected change in the outcome per unit change in the cluster-mean predictor at level 2, without controlling for the level 1 predictor).

When CC is used, as in Equation 1, predictors from different levels are not orthogonal, as the level 1 predictor retains its original information. Therefore, the predictor at level 2 is necessary to unconflate the individual- and cluster-level components. Even when these fixed effects are not fully level-specific, they disaggregate the information in the raw predictor, given that the contextual fixed effect equals the difference of the purely level 2 (between-cluster) and purely level 1 (within-cluster) fixed effects that can be obtained using CWC. This means that both centering techniques lead to likelihood-equivalent models and that contextual fixed effects

can be used to test differences across level-specific fixed effects (Raudenbush & Bryk, 2002; Rights & Sterba, 2023).

Centering is directly related to the problem of conflated fixed and random effects in multilevel analysis. Of the many techniques currently available, this study focuses on the two mentioned above: CC and CWC. These options disaggregate the individual- and cluster-level components of the raw level 1 predictor, either by controlling or removing the influence of one on the other (Hamaker & Muthén, 2020).

Conflated Fixed Effects

Conflated fixed effects, also known as *smushed*, *blended*, or *composite* fixed effects in the multilevel analysis literature, are an uninterpretable weighted average of the within- and between-cluster fixed effects attributable to the individual- and cluster-level components of a level 1 predictor with systematic variance across clusters (Hedeker & Gibbons, 2006; Hoffman, 2015; Hoffman & Walters, 2022; McNeish, 2023; Preacher et al., 2010; Preacher, 2011).

Conflated fixed effects can introduce both statistical and conceptual biases in the analysis, distorting the interpretation of the level 2 component within the level 1 predictor with the level 2 outcomes (Hamaker & Muthén, 2020; Rights & Sterba, 2016; Rights et al., 2020). Additionally, their incorrect standard errors can increase the likelihood of Type I or Type II errors in hypothesis testing (McNeish, 2023). The ambiguity in distinguishing individual- and cluster-level differences also complicates the interpretation of moderation and mediation analyses (Loeys et al., 2018; Preacher, 2011).

The problem of conflated fixed effects and how to avoid them has been extensively studied in the multilevel analysis literature (Hoffman, 2019; Raudenbush & Bryk, 2002; Rights, 2022; Snijders & Bosker, 2012). However, the problem of conflated random slopes has not been

covered with the same intensity (but see Rights & Sterba, 2023, for an exception). Although some work from the 1980s mentioned differences in random slopes according to the centering of the level 1 predictor (Longford, 1989; Raudenbush, 1989; Plewis, 1989), their focus was largely on the interpretation of the model intercept. Both problems have important implications for multilevel analysis, particularly when examining variability across clusters, and they have similar origin and solutions.

Conflated Random Slopes

In multilevel analysis, random slopes serve to capture heterogeneity around the fixed effects of the predictors at level 1 across clusters at level 2 (Hox et al., 2017). By modeling this variability, random slopes offer a flexible and realistic representation of the relationships in clustered data. Ignoring random slopes can lead to model misspecification, biased standard errors, and incorrect inferences about predictor effects (Snijders & Bosker, 2012), although, as we will soon see, even after including them in the model, similar errors can be made.

As discussed above, when we fail to separate the level-specific parts of a raw level 1 predictor, its fixed effect is conflated. Random slopes are not an exception, although their interpretation is different. A conflated or smushed random slope is an uninterpretable blend of both the slope heterogeneity and the intercept heteroscedasticity of the individual- and cluster-level components of the raw level 1 predictor (Rights & Sterba, 2023). For this reason, researchers who do not adequately parameterize random slopes in the model may interpret their variance as a reflection of the heterogeneity of the within-cluster fixed effects across clusters, without realizing that they are also introducing heteroscedasticity across cluster intercepts. This additional intercept heteroscedasticity arising from a conflated random slope is a consequence of

the cluster information contained in the raw level 1 predictor that is not disaggregated or controlled for in the model.

Conflated random slopes can bias the estimation, interpretation, and inferences of the level 2 variances in the MLM, as well as the calculation of the standard errors of the fixed effects and their subsequent Type I or Type II error rates (LaHuis et al., 2020; Rights et al., 2020; Rights & Sterba, 2023).

Relationships Between Centering Techniques and Conflated Random Slopes

When only the CC version of the level 1 predictor is used to estimate its random slope, the within-cluster fixed effect and the intercept variance related to the level 1 and level 2 predictors in the model are forced to have the same pattern of variability across clusters. To understand this constraint, we can go back to the model in Equation 1 and replace the level 1 predictor x_{sc} with an equivalent version that reflects its two components: $x_{sc} = x_{sc} - x_{.c} + x_{.c}$. The purely within-cluster and between-cluster components are $x_{sc} - x_{.c}$ and $x_{.c}$, respectively. This definition comes from the idea of level 1 values as the implicit sum of a cluster score (e.g., its mean) and the specific deviation from that score (Rights & Sterba, 2023). Then, we can include a random slope for x_{sc} in the model to capture cluster differences around its slope γ_{10} (labeled as U_{1c}^{s} with the superscript s to denote its *smushed* parameterization). This model, known as the *conventional random-slope contextual effect MLM* (Rights & Sterba, 2023) is shown in Equation 2:

Level 1

$$y_{sc} = \beta_{0c} + \beta_{1c}([x_{sc} - x_{.c}] + x_{.c}) + e_{sc}$$

$$e_{sc} \sim N(0, \sigma_e^2),$$

Level 2

$$\beta_{0c} = \gamma_{00} + \gamma_{01}^{c}(x_{.c}) + U_{0c}$$

$$\beta_{1c} = \gamma_{10} + U_{1c}^{s}$$

$$\begin{bmatrix} U_{0c} \\ U_{1c}^{s} \end{bmatrix} \sim MVN\left(\begin{bmatrix} 0 \\ 0 \end{bmatrix}, \begin{bmatrix} \tau_{U_0}^{2} & \tau_{01}^{s} \\ \tau_{01}^{s} & \tau_{U_1^s}^{2} \end{bmatrix}\right). \tag{2}$$

The terms in Equation 2 have been explained in Equation 1 and the previous paragraph. Importantly, the variance $\tau_{U_1^s}^{2}$ is capturing the conflated random slopes in the model. For this reason, the covariance with the random intercept τ_{01}^{s} can also have conflation issues. If we check the reduced-form expression of this model in Equation 3, we will see that the conflated random slope U_{1c}^{s} (bolded for emphasis below) is being multiplied twice in the model:

$$y_{sc} = \gamma_{00} + \gamma_{10}([x_{sc} - x_{.c}] + x_{.c}) + \gamma_{01}^{c}(x_{.c}) + \boldsymbol{U_{1c}^{s}}([x_{sc} - x_{.c}] + x_{.c}) + e_{sc}$$

$$y_{sc} = \gamma_{00} + \gamma_{10}(x_{sc} - x_{.c}) + (\gamma_{01} + \gamma_{10}^{c})x_{.c} + \boldsymbol{U_{1c}^{s}}(x_{sc} - x_{.c}) + \boldsymbol{U_{1c}^{s}}(x_{.c}) + e_{sc}, \tag{3}$$

This means that, inadvertently, U_{1c}^{s} is forcing the variability around the within-cluster fixed effect of $x_{sc} - x_{.c}$ and the cluster-specific intercepts across $x_{.c}$ to be equal.

The conventional random-slope contextual effect MLM, widely used in psychology, sociology, and education (Brincks et al., 2016; Hox & Roberts, 2011), is partially conflated (Rights & Sterba, 2023). Its bias depends on the association of the variation of both components of the raw predictor —because of the heterogeneity of the slope and the intercepts were the same, the constriction of the conflated random slope would not be problematic.

As mentioned above, the heteroscedasticity of the cluster-specific intercepts is a consequence of the cluster information that was not separated or controlled in the predictor used to estimate the random slope. If we examine the variance of the cluster-specific intercepts in the

conventional random-slope contextual effect MLM, we will see that their heterogeneity is linked to the cluster-mean predictor $x_{.c}$, as shown in Equation 4:

$$\begin{aligned}
\mathrm{var}(\beta_{0c}|x_{.c}) &= \mathrm{var}(\gamma_{00} + [\gamma_{10} + \gamma_{01}^{c}][x_{.c}] + U_{0c} + U_{1c}^{s}[x_{.c}]) \\
&= \mathrm{var}(U_{0c} + U_{1c}^{s}[x_{.c}]) \\
&= \mathrm{var}(U_{0c}) + 2\mathrm{cov}(U_{1c}^{s}[x_{.c}], U_{0c}) + \mathrm{var}(U_{1c}^{s}[x_{.c}]) \\
&= \mathrm{var}(U_{0c}) + 2\mathrm{cov}(U_{1c}^{s}, U_{0c})[x_{.c}] + \mathrm{var}(U_{1c}^{s})[x_{.c}^{2}] \\
&= \tau_{U_0}^{2} + 2\tau_{01}^{s}(x_{.\mathrm{c}}) + \tau_{U_1^s}^{2}(x_{.c}^{2}). \qquad (4)
\end{aligned}$$

As the fixed effects in the model are constant values, they can be removed when moving from the first step to the second step. At the end of Equation 4, we can see that the model *implies* a heteroscedastic intercept variance, which includes a quadratic function of $x_{.c}$ that is being multiplied by the variance of the conflated random slope U_{1c}^{s}. The quadratic function allows the distribution of the cluster-intercepts to be wider at the extreme values of $x_{.c}$, although in the current MLM the degree of this heterogeneity is forced to be the same as that used to describe the heterogeneity of the outcome variance conditional on the level 1 predictor with the random slope. For completeness, Equation 5 elaborates this problem:

$$\begin{aligned}
\mathrm{var}(y_{sc}|x_{sc}) &= \mathrm{var}(\gamma_{00} + \gamma_{01}^{c}[x_{.c}] + \gamma_{10}[x_{sc}] + U_{0c} + U_{1c}^{s}[x_{sc}] + e_{sc}) \\
&= \mathrm{var}(U_{0c} + U_{1c}^{s}[x_{sc}] + e_{sc}) \\
&= \tau_{U_0}^{2} + 2\tau_{01}^{s}(x_{\mathrm{sc}}) + \tau_{U_1^s}^{2}(x_{sc}^{2}) + \sigma_{e}^{2}. \qquad (5)
\end{aligned}$$

It is worth mentioning that even when level-specific fixed effects are used in the model in Equation 2, this does not solve the conflation of the random slope, given that the between-cluster fixed effect (labeled as γ_{01}^{b}) would not change U_{1c}^{s}.

How to Avoid Conflated Random Slopes

To avoid the estimation of conflated random slopes for level 1 predictors, the same centering options we saw above can be used (CC and CWC), which implies adding a random slope for $x_{.c}$ at level 2. When CC is used, the random slope of the level 2 predictor (labeled as U_{3c}^{c}, whose variance $\tau_{U_3^c}^2$ is the model parameter) captures *differential* heteroscedasticity in the cluster intercepts across the cluster-mean predictor $x_{.c}$. This is because U_{3c}^{c} captures the difference of the level-specific random slopes for the individual- and cluster-level components of the raw level 1 predictor—and because, as we described in the case of fixed effects, this *contextual random slope* is analytically equivalent to the difference between the level-specific random slopes of the predictors at level 1 and level 2 (more information on this can be found in the *Methods* subsection). When CWC is used, the random slope of the level 2 predictor (labeled as U_{3c}^{b}, whose variance $\tau_{U_3^b}^2$ is the model parameter) captures *purely* heteroscedasticity of the cluster intercepts across the cluster-mean predictor $x_{.c}$.

Although the idea of adding a random slope for a level 2 predictor may seem counter-intuitive due to the lack of a higher level across which the fixed effect could vary (Preacher, 2017), its role is to disaggregate sources of variance in the model intercepts (Rights & Sterba, 2016). In addition, as Snijders and Bosker (2012) explain, this is rare but not new in multilevel analysis, as the assumption of homogeneity of variance of the cluster intercepts can be replaced by a function that makes the variance dependent on the values of some predictor at level 2 (whose random slope is what introduces heterogeneity of variance at level 2).

The discussion of conflated random slopes and their consequences is important because these estimates may bias the estimation, interpretation, and inferences of between-cluster differences, leading researchers to erroneous conclusions. In particular, conflated random slopes

may mistakenly imply either a lack of or a false presence of slope heterogeneity (Rights & Sterba, 2023). The extent of these biased inferences depends on the degree of intercept heteroscedasticity across the values of the cluster-mean predictor in the population. For example, if the cluster-specific intercepts in the population do not vary across the cluster-mean predictor (i.e., $\tau^2_{U_3^b} \cong 0$), even when there is heterogeneity around the within-cluster fixed effect across clusters (i.e., $\tau^2_{U_1} > 0$), the variance of the conflated $\tau^2_{U_1^s}$ will be pulled toward 0, increasing the likelihood of concluding there is no slope heterogeneity when there really is (a Type I error).

The alternative error—falsely detecting slope heterogeneity—could be made when the heterogeneity around the within-cluster fixed effect is close to 0 in the population (i.e., $\tau^2_{U_1} \cong 0$), but there is intercept heteroscedasticity across the cluster-mean predictor (i.e., $\tau^2_{U_3^b} > 0$), which can move the conflated $\tau^2_{U_1^s}$ away from 0 instead. In this situation, the conflated random slope would seem to reflect heterogeneity in the within-cluster fixed effect across clusters, when in fact reflects heterogeneity in the cluster-intercepts across values of the cluster-mean predictor.

So far, we have addressed the most relevant aspects of multilevel analysis, level 1 predictors, and conflated fixed and random effects for this study. In addition, we have shown that the parameterization of level 1 predictors is the key to preventing conflated effects in multilevel analysis. This is the background that supports the purpose of the study.

PURPOSE OF THE STUDY

The objective of this study is to analyze the performance of four MLMs to provide unconflated variances for random slopes of level 1 and level 2 predictors under different data conditions inspired by realistic research scenarios. Three of these models have a random slopes parameterization that should adequately capture each source of variability (slope heterogeneity

or heteroscedasticity of the intercepts), whereas the last model serves as a proof-of-concept of the errors to which conflated random effects can lead. Our objective is based on the research question: How do the four multilevel models examined perform and differ in estimating unconflated random slopes across varying data conditions, with respect to bias, power, and standard error accuracy? The evidence from this study can inform decision-making among researchers on how to parameterize their random slopes according to the characteristics of their data.

METHOD

This is a simulation study. To describe the planning and reporting of the data simulation, we used the ADEMP framework proposed by Morris et al. (2019), which is composed of five key steps: Aims, Data-Generating Mechanism (DGM), Methods, Estimands, and Performance Measures. Details on each of these steps are provided below.

Aims

The aim of the simulation was to provide evidence on the performance of three MLMs to recover unconflated random slope variances (the fourth MLM, the conventional random-slope contextual effect MLM, had conflated random slopes). This required the analysis of different performance measures of the model results (i.e., bias, power, and empirical SE), which are described in detail in the Performance Measures subsection. These performance measures assessed the capability of the MLMs to recover the *true* variances of the random slopes, which should capture (purely or unconflated) slope heterogeneity around the fixed effect of the level 1 predictor across clusters.

The simulation will help in finding data conditions in which variance recovery would not be possible or reliable, which can contribute to understanding when a model parameterization

may fail to disaggregate random slopes and alternative options should be considered—such failure can be related to the complexity of the model or the lack of information to estimate the parameters, regardless of the unconflated parameterization of the random slopes.

Data-Generating Mechanism

Data were simulated with the DGM shown in Equation 6:

Level 1

$$y_{sc} = \beta_{0c} + \beta_{2c}(x_{sc} - x_{.c}) + e_{sc}$$

$$e_{sc} \sim N(0, \sigma_e^2),$$

Level 2

$$\beta_{0c} = \gamma_{00} + \gamma_{01}^b(x_{.c}) + U_{0c} + U_{3c}^b(x_{.c})$$

$$\beta_{2c} = \gamma_{20} + U_{2c}$$

$$\begin{bmatrix} U_{0c} \\ U_{2c} \\ U_{3c}^b \end{bmatrix} \sim MVN\left(\begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} \tau_{U_0}^2 & \tau_{02} & \tau_{03}^b \\ \tau_{02} & \tau_{U_2}^2 & \tau_{23}^b \\ \tau_{03}^b & \tau_{23}^b & \tau_{U_3^b}^2 \end{bmatrix}\right), \tag{6}$$

in which a continuous outcome y_{sc} for student s ($s = 1, \ldots, S$) in classroom c ($c = 1, \ldots, C$) was predicted by the CWC version of the level 1 predictor ($x_{sc} - x_{.c}$) and the CC version of the cluster-mean predictor ($x_{.c}$ centered at 0), whose fixed effects γ_{10} and γ_{01}^b are the within- and between-cluster fixed effects, respectively. In addition, the predictors each had a random slope at level 2: U_{2c}, that captured purely slope heterogeneity around the within-cluster fixed effect across clusters, and U_{3c}^b, that captured purely heteroscedasticity of the cluster-intercepts across the cluster-mean predictor. The notation of the models in this section necessarily differs from that shown above, as there are more random slopes to represent. For consistency with the notation of

the analysis models below, the level 1 equation has skipped over β_{1c}, which will be reserved for use with the CC version of the level 1 predictor instead.

The level 1 residuals e_{sc} in the DGM were assumed to be independently and normally distributed with a mean of zero and a constant variance σ_e^2. The level 2 residuals were assumed to be independently and multivariate normally distributed with a vector of means equals to 0 and an estimated variance–covariance matrix including all possible covariances.

The parameterization of the fixed and random effects in the DGM allowed us to set level-specific true values during the data simulation. The conditions for the generation of the outcome y_{sc} are described in Table 1.

Table 1. Factors and Levels of the Data-Generating Mechanism.

Factor	**Notation**	**True Values or Levels**
Number of replications per combination	n_{sim}	1,000
Sample Size		
Level 1	$L1_n$	5 and 30
Level 2	$L2_n$	25 and 50
Predictors		
Level 1 (CWC version)	$x_{sc} - x_{.c}$	$\mu = 0$; $\sigma = \sqrt{\text{Variance of } x_{sc} - x_{.c}}$
Level 2 (CC version)	$x_{.c}$	$\mu = 0$; $\sigma = \sqrt{\text{Variance of } x_{.c}}$
Fixed Effects		
Intercept	γ_{00}	10
Purely within-cluster	γ_{10}	1
Purely between-cluster	γ_{01}^b	3
Intraclass Correlation Index		
ICC of x_{sc}	$ICC_{x_{sc}}$	0.5

ICC of y_{sc} (conditional on x_{sc} = 0 and $x_{.c}$ = 0)	$ICC_{y_{sc}}$	0.25
Variances		
Total variance of y_{sc}	$\sigma^2_{y_{sc}}$	16
Total variance of x_{sc}	$\sigma^2_{x_{sc}}$	4
Variance of $x_{sc} - x_{.c}$	$\sigma^2_{x_{sc}} L1$	$\sigma^2_{x_{sc}} * (1 - ICC_{x_{sc}})$
Variance of $x_{.c}$	$\sigma^2_{x_{sc}} L2$	$\sigma^2_{x_{sc}} * ICC_{x_{sc}}$
Random slope reliability	SR	0.2 and 0.6
Residual variance at level 1	σ^2_e	$\sigma^2_{y_{sc}} * (1 - ICC_{y_{sc}})$
Random intercept variance	$\tau^2_{U_0}$	$\sigma^2_{y_{sc}} * ICC_{y_{sc}}$
Random slope variance for the level 1 predictor ($x_{sc} - x_{.c}$)	$\tau^2_{U_2}$	$\tau^2_{U_2} = \dfrac{SR + \dfrac{\sigma^2_e}{L1_n * \sigma^2_{x_{sc}}}}{1 - SR}$
Random slope variance for the level 2 predictor ($x_{.c}$)	$\tau^2_{U_3^b}$	$\tau^2_{U_2}$* (0.5, 1.0, and 1.5), which represent 50% less or more variability with respect to $\tau^2_{U_2}$
Correlation between U_{0c} and U_{2c}	τ_{02}	.1
Correlation between U_{0c} and U^b_{3c}	τ^b_{03}	.1
Correlation between U_{2c} and U^b_{3c}	τ^b_{23}	.1
Outcome	y_{sc}	Based on the model in Equation 6

Note. CWC = Centering–within–cluster; CC = Constant–centering.

The number of replications for each of the 24 unique combinations of factor levels was 1000. The sample sizes at level 1 and level 2 varied according to practical and theoretical limits. We know that the stability of a model estimates decreases as the sample size at either level decreases, so the lower bounds were chosen to represent a situation of theoretically minimal conditions under which it should be possible to recover the true random slope variances. The minimum number of clusters was 25, which is below some common recommendations found in

the multilevel analysis literature (McNeish & Stapleton, 2016; Snijders & Bosker, 2012), but it is a realistic scenario. Regarding level 1 sample size, we chose values of 5 and 30 to analyze how the number of units per cluster may affect the performance of the MLMs under study, given that this data condition may affect the information available in the models to estimate parameters such as the random slopes (Hox et al., 2017; Snijders, 2005).

The predictors at level 1 and level 2 were created taking as reference a raw predictor x_{sc} with systematic variance across clusters $ICC_{x_{sc}} = .5$. Therefore, in

Table 1 the notation x_{sc} refers to a predictor that contains within- and between-cluster information, whereas the notation $x_{sc} - x_{.c}$ and $x_{.c}$ refer to the purely within- and between-cluster components of x_{sc}, respectively. These level-specific variables were used to estimate the unconflated fixed and random effects in the DGM that should be recovered by the models. The ICC of the outcome y_{sc} was fixed at .25 so as not to unnecessarily increase the complexity of the analysis and the number of factors.

To define the true values of the random slope variance for the level 1 predictor ($\tau^2_{U_2}$) we used reverse engineering. In practice, the variance components and level-specific sample sizes can be used to check the reliability around the random effects in the model (Hox et al., 2017). Given the information, we could use Equation 7 to calculate the reliability of the random slope U_{2c} (adapted from Willett, 1989), as slope reliability (SR):

$$\text{SR} = \frac{\tau^2_{U_2}}{\tau^2_{U_2} + \frac{\sigma^2_e}{L1n * \sigma^2_{x_{SC}} L1}}, \tag{7}$$

whose terms were defined in Table 1. By manipulating this formula, $\tau^2_{U_2}$ was created from specific values for SR in the DGM, as shown in Equation 8:

$$\tau^2_{U_2} = \frac{SR + \frac{\sigma^2_e}{L1 n^* \sigma^2_{x_{SC}} L1}}{1 - SR}. \quad (8)$$

The SR was set to either 0.2 or 0.6, reflecting realistic scenarios in which the random slope variance is either weakly or moderately identifiable across clusters. This reliability value quantifies the proportion of observed variability in estimated slopes that can be attributed to true differences in slopes between clusters, rather than sampling error.

Defining true values for $\tau^2_{U_3^b}$ was challenging. Although a similar approach as with $\tau^2_{U_2}$ could have been used, we opted to vary $\tau^2_{U_3^b}$ as a function of $\tau^2_{U_2}$. This decision was based on how the sample size and variances of level 1 and level 2 predictors differed across conditions, which makes an extension of Equation 8 difficult to defend. For these reasons, the random slope variance for the level 2 predictor ($\tau^2_{U_3^b}$) increases or decreases between 0.5 and 1.5 relative to $\tau^2_{U_2}$.

Estimands

The estimands in the simulation were the variances of the random slopes and the SE of the fixed effects at level 1 and level 2. The performance measures captured the precision and variability in the estimation of these estimands across the replications and combinations of factor levels for the MLMs examined.

Methods

In this study, we analyzed the performance of four MLMs to recover the unconflated random slope variances created in the DGM. The first three MLMs (shown in Equations 9, 10, and 11) were recommended by Rights and Sterba (2023) to avoid the estimation of conflated random slopes, and the last model is the conventional random-slope contextual effect MLM

(shown in Equation 2), whose random slope remains conflated. Table 2 shows each MLM in detail. Importantly, the parametrization of the fixed effects is the same across models—within-cluster plus contextual fixed effects—on purpose to focus on the parameterization of the random slopes. Furthermore, in Models 2 and 3 we reserved U_{1c} for the random slope of the CC predictor at level 1, U_{3c}^{b} for the random slope of the cluster-mean predictor paired with a CWC predictor at level 1, and U_{3c}^{c} for the random slope of the cluster-mean predictor paired with a CC predictor at level 1.

For greater transparency, we simplified the names of these MLMs relative to the longer terms originally used by Rights and Sterba (2023). However, in this document we will often refer to these models according to their characteristic random slope. Thus, Model 1 will be referred to as the *within-only* model, Model 2 as the *within–between* model, Model 3 as the *within–contextual* model, and Model 4 as the *random-conflated* model.

Table 2. Models Examined.

Model 1: **Within-cluster random slope (within-only)**

Level 1

$$y_{sc} = \beta_{0c} + \beta_{1c}(x_{sc}) + \beta_{2c}(x_{sc} - x_{.c}) + e_{sc}$$
$$e_{sc} \sim N(0, \sigma_e^2),$$

Level 2

$$\beta_{0c} = \gamma_{00} + \gamma_{01}^{c}(x_{.c}) + U_{0c}$$
$$\beta_{1c} = \gamma_{10}$$
$$\beta_{2c} = U_{2c}$$
$$\begin{bmatrix} U_{0c} \\ U_{2c} \end{bmatrix} \sim MVN\left(\begin{bmatrix} 0 \\ 0 \end{bmatrix}, \begin{bmatrix} \tau_{U_0}^2 & \tau_{02} \\ \tau_{02} & \tau_{U_2}^2 \end{bmatrix}\right) \quad (9)$$

Model 2: **Within-cluster and between-cluster random slopes (within–between)**

Level 1

$$y_{sc} = \beta_{0c} + \beta_{1c}(x_{sc}) + \beta_{2c}(x_{sc} - x_{.c}) + e_{sc}$$
$$e_{sc} \sim N(0, \sigma_e^2),$$

Level 2

$$\beta_{0c} = \gamma_{00} + \gamma_{01}^{c}(x_{.c}) + U_{0c} + U_{3c}^{b}(x_{.c})$$
$$\beta_{1c} = \gamma_{10}$$
$$\beta_{2c} = U_{2c}$$
$$\begin{bmatrix} U_{0c} \\ U_{2c} \\ U_{3c}^{b} \end{bmatrix} \sim MVN\left(\begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} \tau_{U_0}^2 & \tau_{02} & \tau_{03}^{b} \\ \tau_{02} & \tau_{U_2}^2 & \tau_{23}^{b} \\ \tau_{03}^{b} & \tau_{23}^{b} & \tau_{U_3^b}^2 \end{bmatrix} \right) \quad (10)$$

Model 3: **Within-cluster and contextual random slopes (within–contextual)**

Level 1

$$y_{sc} = \beta_{0c} + \beta_{1c}(x_{sc}) + e_{sc}$$
$$e_{sc} \sim N(0, \sigma_e^2),$$

Level 2

$$\beta_{0c} = \gamma_{00} + \gamma_{01}^{c}(x_{.c}) + U_{0c} + U_{3c}^{c}(x_{.c})$$
$$\beta_{1c} = \gamma_{10} + U_{1c}$$
$$\begin{bmatrix} U_{0c} \\ U_{1c} \\ U_{3c}^{c} \end{bmatrix} \sim MVN\left(\begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} \tau_{U_0}^2 & \tau_{01} & \tau_{03}^{c} \\ \tau_{01} & \tau_{U_1}^2 & \tau_{13}^{c} \\ \tau_{03}^{c} & \tau_{13}^{c} & \tau_{U_3^c}^2 \end{bmatrix} \right) \quad (11)$$

Model 4: **Conventional random-slope contextual effect MLM (random-conflated)**

Level 1

$$y_{sc} = \beta_{0c} + \beta_{1c}([x_{sc} - x_{.c}] + x_{.c}) + e_{sc}$$
$$e_{sc} \sim N(0, \sigma_e^2),$$

Level 2

$$\beta_{0c} = \gamma_{00} + \gamma_{01}^{c}(x_{.c}) + U_{0c}$$
$$\beta_{1c} = \gamma_{10} + U_{1c}^{s}$$
$$\begin{bmatrix} U_{0c} \\ U_{1c}^{s} \end{bmatrix} \sim MVN\left(\begin{bmatrix} 0 \\ 0 \end{bmatrix}, \begin{bmatrix} \tau_{U_0}^2 & \tau_{01}^{s} \\ \tau_{01}^{s} & \tau_{U_1^s}^2 \end{bmatrix}\right). \quad (2)$$

The within-only model exclusively adds heterogeneity around the within-cluster fixed effect of the level 1 predictor across clusters. Although it is uncommon to include different versions of the level 1 predictor in the fixed and random parts of the model, this so-called "hybrid" parameterization disaggregates the random slope variance $\tau_{U_1^s}^2$ while maintaining a contextual interpretation of the level 2 fixed effect of $x_{.c}$. One thing to consider is that this model assumes intercept homoscedasticity, given that a term to capture intercept heteroscedasticity across the cluster-mean predictor has not been included (i.e., $\tau_{U_3^b}^2 = 0$). Therefore, this model is a good choice for researchers who are only interested in capturing slope heterogeneity across clusters or who did not find systematic variance in their level 1 predictors across clusters.

The within–between and within–contextual models add slope heterogeneity and intercept heteroscedasticity terms to the analysis. In these models, U_{1c} and U_{2c} capture slope heterogeneity around the fixed effect of the level 1 predictor across clusters, whereas U_{3c}^{b} and U_{3c}^{c} serve to capture intercept heteroscedasticity associated with the cluster-mean predictor, reflecting pure or differential heteroscedasticity, respectively. This difference in the interpretation of the random slope of the cluster-mean predictor in the within–between and within–contextual models is a

consequence of the centering approach, CWC or CC, although the corresponding random slope variances under both approaches are analytically equivalent. Specifically, the variance of U_{3c}^{b} in the within–between model (using CWC) is analytically equivalent to the sum of the variances of U_{1c} and U_{3c}^{c} from the within–contextual model (using CC), whereas the variance of U_{3c}^{c} in the within–contextual is analytically equivalent to the difference of the variance of U_{3c}^{b} and U_{2c} from the within–between model. Notably, both operations involve adding twice the covariance between the random slopes, which can appear counterintuitive when moving from the within–between model to the within–contextual model. Additional details on this analytical equivalence are provided in the Appendix.

The random-conflated model served as a comparison criterion to show the potential errors that can be made when not disaggregating random slopes. This model, particularly popular in social sciences (Rights & Sterba, 2023), imposes restrictions on the variance components at level 2 that are not usually found in real data, which, as we have described above, can lead to substantive and statistical problems.

In the four models examined, a relevant difference is the interpretation of the variance of the random slope, which is not constant because the predictors with random slope vary across models—and the variance of the random intercept is conditioned on the values of the predictors with random slope equal to 0. In the within-only model, the random intercept variance is conditioned on units whose level 1 predictor values are equal to their cluster mean, whereas in the within–between model the random intercept variance is conditioned on units whose level 1 predictor values are equal to their cluster mean within average clusters, in the within–contextual model is conditioned on units whose level 1 predictor values are 0 within average clusters, and in the random conflated model is conditioned on average units. The idea of *average* units and

clusters comes from the way in which the variables were constructed at level 1 and level 2: both are normally distributed variables with a mean of 0 and a true variance. This means that 0 in both variables is close to the mean or the average.

A summary of the main features of each MLM examined is included in Table 3.

Table 3. Main Features of the Models Examined.

Model	**Random slopes**	**What is being added to the model?**	**Conflated slope?**	**Key assumption**
Within-only	$U_{2c}(x_{sc} - x_{.c})$	Purely slope heterogeneity	No	Intercept homoscedasticity $\tau^2_{U_3^b} = 0$
Within–between	$U_{2c}(x_{sc} - x_{.c})$, $U^b_{3c}(x_{.c})$	Slope heterogeneity and intercept heteroscedasticity	No	
Within–contextual	$U_{1c}(x_{sc})$, $U^c_{3c}(x_{.c})$	Slope heterogeneity and differential intercept heteroscedasticity	No	
Random-conflated	$U^s_{1c}(x_{sc})$	Uninterpretable blend of slope heterogeneity and intercept heteroscedasticity	Conflated random slope	Equal random slopes $\tau^2_{U_2} = \tau^2_{U_3^b}$

The simulation and all the analyses were conducted in the R environment v. 4.3.1 with the help of the Argon High-Performance Computing (HPC) system of the University of Iowa. The MLMs were estimated using restricted maximum likelihood (REML) given the emphasis on the variance components of each model, with the *lmer* function in the `lme4` package (Bates et al., 2015). Relative model fit assessment—used to determine whether the parameters recovered were statistically significant—was performed via likelihood ratio tests (-2ΔLL), also known as

deviance tests, in which the difference in log-likelihood when comparing nested models is multiplied by −2 and evaluated against a mixture χ^2 distribution (50:50 mixture of χ^2 with degrees of freedom based on the difference in the number of model parameters) as reference.

Performance Measures

The performance of the MLMs analyzed is evaluated by performance measures, which are numerical indices that provide information about the accuracy and bias of simulated results (Morris et al., 2019). Table 4 contains the operationalization of all the performance measures included in the evaluation of the models.

Table 4. Key Performance Measures.

Measure	**Definition**	**Estimate**
Bias	$E[\hat{\theta}] - \theta$	$\frac{1}{n_{sim}} \sum_{i=1}^{n_{sim}} \hat{\theta}_i - \theta$
Empirical SE	$\sqrt{Var(\hat{\theta})}$	$\sqrt{\frac{1}{n_{sim}-1} \sum_{i=1}^{n_{sim}} (\hat{\theta}_i - \bar{\theta})^2}$
Power	$\Pr(p_i \leq \alpha)$	$\frac{1}{n_{sim}} \sum_{i=1}^{n_{sim}} 1(p_i \leq \alpha)$

Note. θ corresponds to the true value of the estimand of interest; $\hat{\theta}_i$ corresponds to the estimated value of the estimand of interest; $\bar{\theta}$ corresponds to the mean of the estimated values of the estimand of interest.

Bias is a substantial performance measure for evaluating our model results that quantifies the systematic deviation of the estimates from the true values (Banack et al., 2021). When we evaluate the bias of the estimates, we are evaluating whether we are getting the right answer on average across multiple repeated samples (Carsey & Harden, 2013). For example, the estimator

$\hat{\theta}$ of the true value of the random slope variance of the predictor at level 2 ($\tau^2_{U_3^b}$ labeled as θ) is unbiased if $E(\hat{\theta}) = \theta$. This performance measure served to assess whether the estimates of the models were close enough on average to the true value of the estimands across replications.

The empirical standard error (empSE) was used to evaluate the precision of the estimators across replications (if $\hat{\theta}$ was an efficient estimator of θ). Following Morris et al. (2019), the empirical SE is defined as the long-run standard deviation of the estimates $\widehat{\theta_l}$ over the number of replications. This quantity reflects how much the estimates fluctuate from sample to sample, providing insight into the efficiency of the estimator under a given model. For fixed effects, a comparison of empirical SEs can inform how consistently a model recovers these parameters across simulated samples. Lower empirical SEs suggest higher precision in estimation, which may influence the selection of model specifications to get accurate inferences about fixed effects.

Finally, as in Rights and Sterba (2023), we evaluated the power of each model to detect the presence of the random slope variance of the level 2 predictor, operationalized as the proportion of replications in which a likelihood ratio test rejected the null hypothesis of zero variance. This performance measure addresses whether the models can reliably detect a non-zero variance signal under the simulated conditions.

Procedure

The results of the simulation were analyzed in two phases using different regression models and effect sizes to identify the most prominent factors for the analysis of conflated random slopes. First, the convergence of each model across data conditions was analyzed using a binary regression model with five factors: model, sample size at level 1 and level 2, reliability of the random slope, and random slope variance for the predictor at level 2. All the main fixed

effects and possible interactions were included in the model described conceptually in Equation 12:

$$
\begin{aligned}
logit[\Pr(convergence = 1)] = \; & Model + \\
& Sample\ Size\ at\ Level\ 1 + \\
& Sample\ Size\ at\ Level\ 2 + \\
& Random\ Slope\ Reliability + \\
& Random\ Slope\ Variance\ Predictor\ at\ Level\ 2 + \\
& Interactions
\end{aligned} \tag{12}
$$

The model in Equation 12 uses a conditional Bernoulli distribution for the outcome and the logit link function to predict the probability of convergence equals one. Because the Bernoulli distribution does not have an estimated variance, calculating effect sizes such as η^2 for the model factors is not straightforward, so we have used likelihood ratio tests ($-2\Delta\mathrm{LL}$) to assess the relevance of each factor in predicting the convergence of the models using a nominal $\alpha < .01$.

The binary logistic regression model was useful for analyzing the effect of the complexity of the models under study on the recovery of random slope variances across the combination of factor levels (i.e., data conditions). Additional bivariate correlations were calculated to analyze how convergence ratios, design factors, and true values of certain parameters were associated across models.

Second, the performance measures described in Table 4 for each parameter of interest were analyzed in two datasets according to models that converged individually in each replication (labeled as *convergence by models*), and results from models that converged in replications without problems in any model (labeled as *convergence by replications*). Although we originally considered focusing only on the second dataset to maintain a constant number of

replications across models, analysis of the first dataset is important to examine the possible effect of model complexity on the sampling distribution of the performance measures. By focusing only on the results of models from replications without convergence problems, we could introduce selection bias that distorts our conclusions about how data conditions are associated with model performance in terms of bias, power, and empirical SE for the parameters of interest.

Parameter bias was analyzed using traditional ANOVA regression models (e.g., five-way ANOVA models). The relevance of each factor and their interactions was evaluated using partial η^2 effect sizes calculated with the `effectsize` package functions (Ben-Shachar et al., 2020). Bias was analyzed in both the convergence by models and convergence by replications datasets.

RESULTS

For each of the 24 combinations of factor levels, 1,000 datasets were created using the reference values mentioned in Table 3.

Table 3. For each dataset, the four models described in

Table 2 were fitted, in addition to a null model with a random intercept that was used to perform the deviance test ($-2\Delta\text{LL}$) that assessed the statistical significance of the random slope of the level 1 predictor in the other models. The results of the models were saved in a dataset (n = 120,000) and analyzed in terms of convergence and according to the performance measures described in Table 4.

Model Convergence

Model convergence was evaluated using three diagnostic criteria (based on the tools available in the `lme4` package): singularity, negative eigenvalues, and gradient-based tolerance. Singularity refers to a condition in which estimated variance components or correlation parameters approach their boundary values (e.g., variances ≈ 0, correlations ≈ ±1), indicating that

one or more random effects may be poorly identified. Negative eigenvalues are related to the Hessian matrix in the model (i.e., the matrix of the second derivatives) and suggest that the optimization routine did not successfully locate a proper maximum of the likelihood function, implying that some estimates were unstable or unreliable. Gradient-based tolerance failures occur when the maximum absolute value of the gradient (i.e., the first derivative of the likelihood function with respect to any parameter) exceeds the convergence threshold used by `lme4` (0.002). This indicates that the optimization stopped near—but not exactly at—a stationary point, and thus the algorithm may not have fully converged. Models that triggered any of these warnings were flagged as not having converged in a binary outcome (0 = the model did not converge, 1 = the model converged).

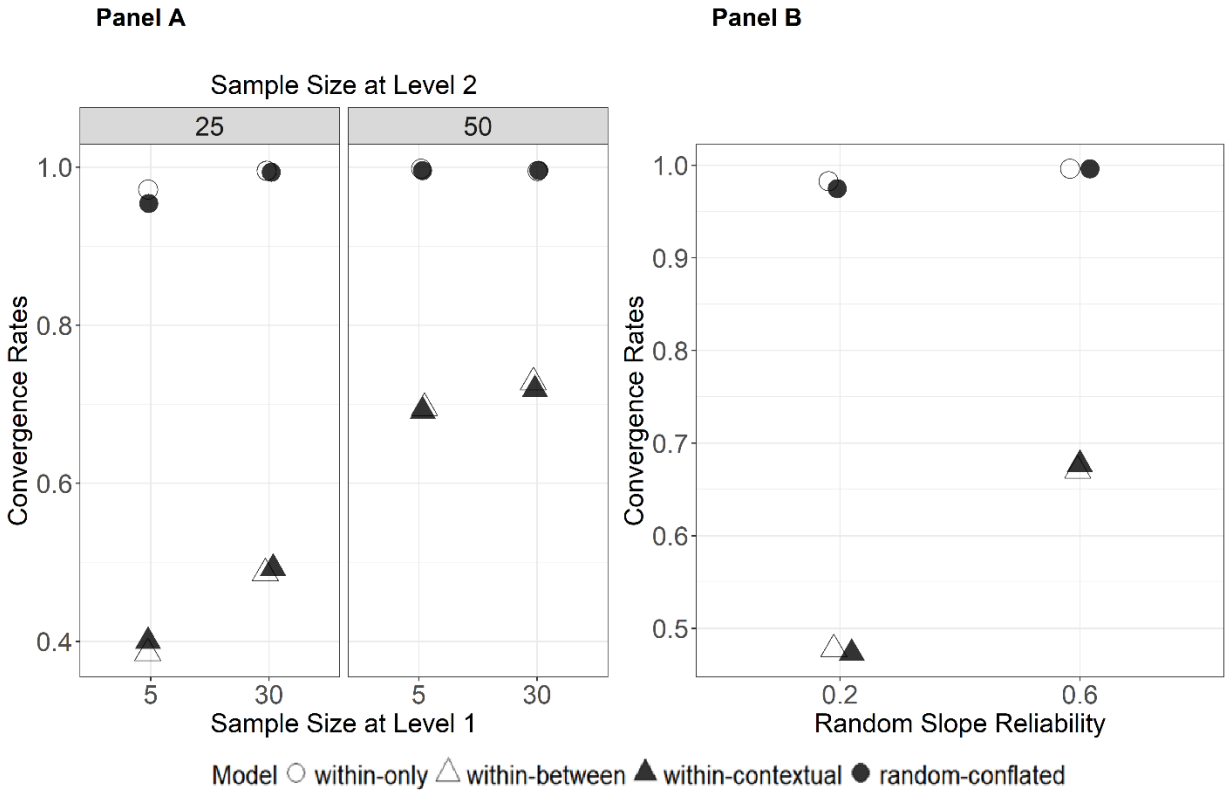
In general, 16% of the simulation results had singularity problems, whereas 2% had problems with negative eigenvalues or gradient-based tolerance. Therefore, around 18% of the results had some convergence issues. Next, the convergence outcome was aggregated through the levels of the factors using means, creating convergence rates for each model across data combinations. Specifically, convergence rates were .99 for the within-only model, .57 for the within–between model, .56 for the within–contextual model, and .98 for the random-conflated model.

The five-way binary logistic regression model in Equation 12 was fitted using the simulation results and the relevance of each factor and its interactions was examined based on $-2\Delta\mathrm{LL}$ tests. These tests were performed using the Type III or marginal sum of squares. The most prominent factors for model convergence were the three-way interaction of model with the sample size at level 1 and level 2, $-2\Delta\mathrm{LL}(4) = 14.78$, $p < .01$, and the two-way interaction of

model with the reliability used to create the true variance of the random slope of the level 1 predictor, $-2\Delta LL(4) = 32.94$, $p < .01$.

When analyzing convergence rates according to model and sample size at level 1 and level 2, the smallest rates were found in the models with a random slope for the level 2 predictor (i.e., the within–between and within–contextual models). While models with only a random slope for the level 1 predictor achieved convergence rates ranging from .95 to .99 across samples size combinations (i.e., the within-only and random-conflated models, respectively), the within–between and within–contextual models achieved convergence ratios between .38 and .73 under these data conditions. These results are shown in Panel A of Figure 1, in which we added a small amount of random noise to the position of the data points to reduce overlap, given that very similar convergence rates between some pairs of models were found. Moreover, the white and black circles represent the within-only and random-conflated models, respectively, whereas the white and black triangles indicate the within–between and within–contextual models, respectively.

Figure 1. Convergence Rates of Models.



Note. Jitter was added to reduce overlap; Panel A depicts the convergence ratios of the models in Table 2 according to sample sizes at level 1 and level 2; Panel B depicts the convergence ratios of the models according to the reliability used to create the variance of the random slope of the level 1 predictor.

The lowest convergence rates for the within–between and within–contextual models were found in the smallest sample sizes at level 1 and level 2 (.38 and .40, respectively). On the contrary, their highest convergence rates were found in the largest sample size combinations (.73 and .72, respectively). These results highlight the importance of sample sizes for the convergence of models with a random slope for a predictor at level 2, which, although adequate for avoiding conflated random slopes, comes at a cost for model estimation. Furthermore, as shown in Panel

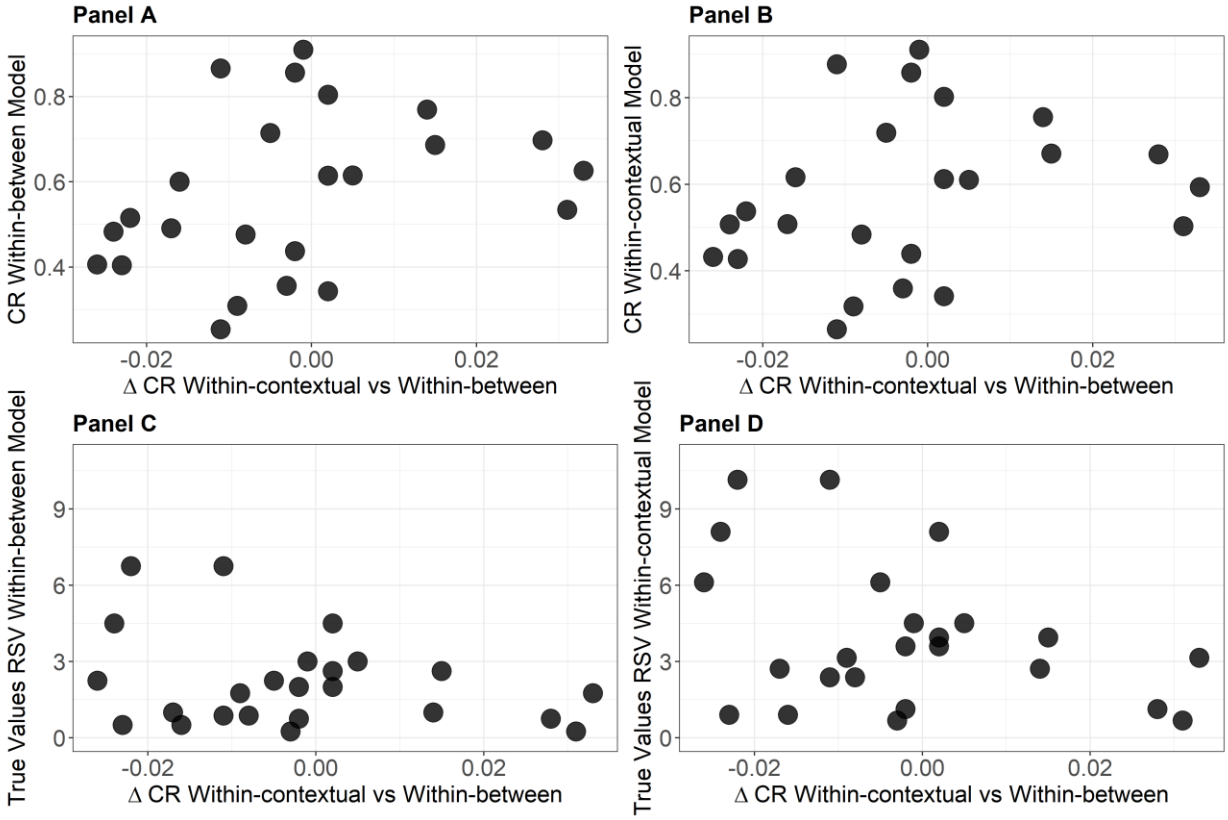
A, simpler models (i.e., the within-only and random-conflated models) achieved convergence rates above .97 in all sample size combinations.

More convergence problems for the within–between and within–contextual models were found across the random slope reliability levels. While the within-only and random-conflated models achieved convergence rates above .97 at random slope reliability 0.2 and 0.6, the within–between and within–contextual models achieved around .47 and .67 convergence rates across these conditions. These results are shown in Panel B of Figure 1. Once again, the convergence of models with a random slope for the predictor at level 2 was low, highlighting the importance of information on between-cluster differences for estimating complex models.

Although the convergence rates of the within–between and within–contextual models were very similar, we calculated their difference (ΔCR) across the 24 factor level combinations to analyze the possible effect of the parameterization of the random slope on the success/failure of each model. Figure 2 shows the ΔCR in which negative values mean that the within–contextual model achieved better convergence and positive values mean that the within–between was superior.

At first glance, in Panels A and B of Figure 2, the difference in convergence slightly favored the within–contextual model over the within–between model (there were more negative than positive values) and seemed weakly correlated to the convergence rate of each model (e.g., the correlations of ΔCR with the convergence rates of these models were .34 and .25, respectively). In addition, the within–contextual model outperformed the within–between in combinations with low individual convergence rates (lower left corner of Panels A and B), although, in general, these differences were less than 0.05.

Figure 2. Differences in Convergence Rates of Within–Between and Within–Contextual Models.



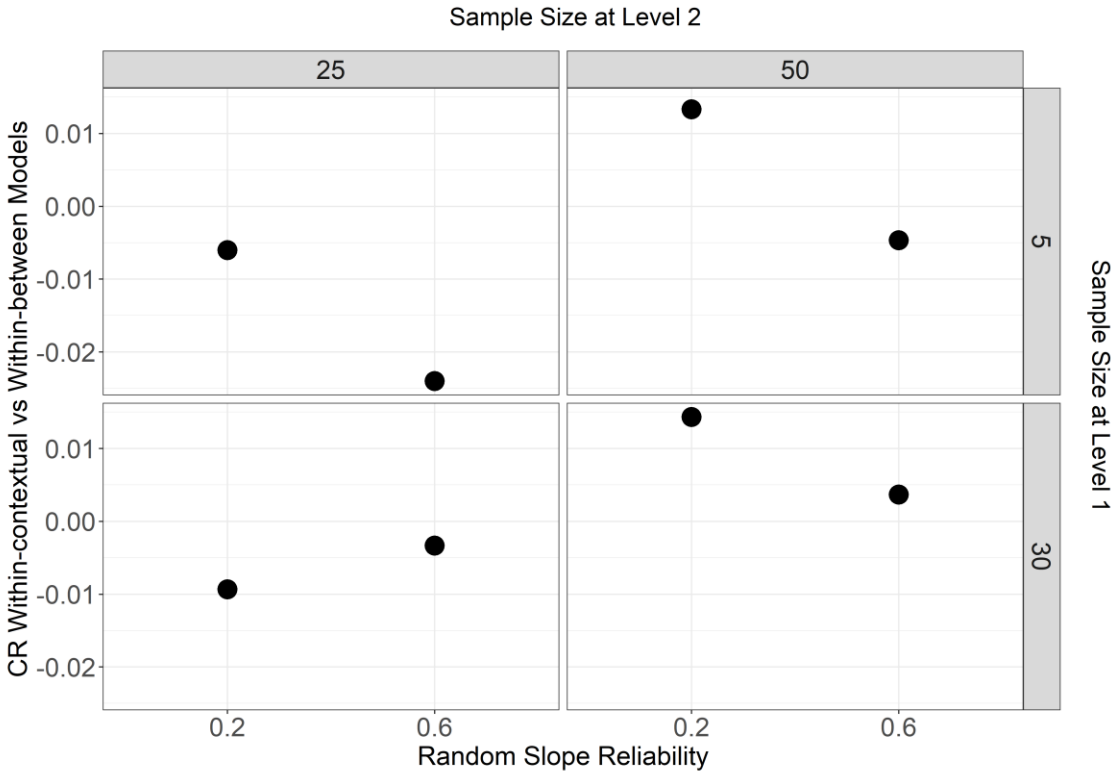
Note. CR = Convergence rate; The difference in convergence rates (ΔCR) was calculated as the average convergence of within–between model minus the average convergence of within–contextual model; Panels A and B depict the relationship of the difference in convergence rates with the individual convergence rate of each model, whereas panels C and D depict the relationship of the difference in convergence rates with the true variance of the random slope characteristic for each model; RSV = Random slope variance for the predictor at level 2.

Panels C and D of Figure 2 show the relationship of ΔCR with the true values for the variance of the random slope of the level 2 predictor in the models to analyze whether model performance could be related to the size of these parameters. Based on these panels, the within–contextual model achieved better convergence as the reference variance increased, whereas the

within–between model had better convergence in combinations with small variances. In a nutshell, the within–contextual parameterization appears to be slightly superior to the within–between parameterization, especially when the random slope variance of the level 2 predictor is small.

To complete this subsection, we examined the association of ΔCR with the factors just analyzed above (i.e., sample size at level 1 and level 2 and random slope reliability). Figure 3 shows these relationships. The within–between model outperformed the within–contextual model when the sample size at level 2 was large (captured as positive ΔCR), whereas the opposite performance was found in the poorest sample size conditions (captured as negative ΔCR).

Figure 3. Difference in Convergence Rates of Within–Between and Within–Contextual Models.



Note. CR = Convergence rates.

Performance Measures: Fixed and Random Effects Bias

The bias of the fixed and random effects, calculated as the difference of the estimator and the true value of each parameter (i.e., $\widehat{\theta}_i - \theta$), was analyzed using general linear regression models with the same factors as the model in Equation 12. Specifically, the five-way ANOVA model used in this subsection to analyze the bias of parameter i ($i = 1, \ldots, I$) was:

$$
\begin{aligned}
Bias_i = \; & Model + \\
& Sample\ Size\ at\ Level\ 1 + \\
& Sample\ Size\ at\ Level\ 2 + \\
& Random\ Slope\ Reliability + \\
& Random\ Slope\ Variance\ Predictor\ at\ Level\ 2 + \\
& Interactions + e_i \\
& e_i \sim N(0, \sigma_e^2)
\end{aligned} \tag{13}
$$

To assess the relevance of the factors, partial η^2 effect sizes using Type III or marginal sum of squares were calculated. As mentioned above, the bias was analyzed in two datasets: one contained the results of each model that converged through replications (similar to assuming pairwise deletion, labeled as the *convergence by model* dataset) and the other contained the results of models that converged jointly through replications (similar to assuming listwise deletion, labeled as the *convergence by replication* dataset). The comparison of results across both datasets may indicate a selection bias that would not be otherwise noticed. The dataset filtered by model convergence contained 82.46% of the original results, and the dataset filtered by replication contained 50.23% of the original results.

In both the convergence by model and convergence by replication datasets, the bias means for the fixed intercept and within-cluster fixed effect were very small (< |0.01|), and no

salient effect sizes were found in the regression models ($\eta^2 < 0.01$). For the between-cluster and contextual fixed effects, the bias means were around −0.19 across the filtered datasets, indicating a consistent underestimation across results. The true between-cluster fixed effect was 3 and the true contextual fixed effect was 2 (which is the difference of the true between-cluster fixed effect = 3 and the true within-cluster fixed effect = 1). In both the convergence by model and convergence by replication datasets, the mean estimated between-cluster fixed slope was around 2.81 (instead of 3), whereas the mean estimated contextual fixed slope was close to 1.81 (instead of 2).

With respect to the effect sizes of the factors in the models for bias of between-cluster and contextual fixed effects, only the η^2 for sample size at level 1 was salient ($\eta^2 \approx 0.05$) across datasets. As shown in As might be expected, the bias means for these fixed effects improved markedly when the sample size at level 1 increased.

Table 5, the bias means for the between-cluster and contextual fixed effects were close to −0.33 in small sample sizes at level 1 across datasets, which means that these fixed slopes were consistently underestimated, even though there was a large difference in the amount of data lost due to convergence issues. As might be expected, the bias means for these fixed effects improved markedly when the sample size at level 1 increased.

Table 5. Bias Means for the Between-cluster and Contextual Fixed Effects.

Factors	Convergence by Model		Convergence by Replication	
	Bias mean	Proportion results	Bias mean	Proportion results
Between-cluster Fixed Effect				
- Sample Size at Level 1 = 5	−0.33	.81	−0.33	.47
- Sample Size at Level 1 = 30	−0.06	.84	−0.06	.53

Contextual Fixed Effect				
- Sample Size at Level 1 = 5	−0.33	.81	−0.33	.47
- Sample Size at Level 1 = 30	−0.06	.84	−0.06	.53

Note. The true values for the between-cluster and contextual fixed effects were 3 and 2, respectively.

The prominent factors in the model for random intercept variance bias varied slightly across datasets. In the model fitted using the results from convergence by model, the two-way interaction of model with the random slope reliability showed a salient η^2 = 0.04, together with the random slope variance for the level 2 predictor and the sample size at level 1, whose η^2 were 0.18 and 0.19, respectively. Table 6 shows these results. Interestingly, all bias means were positive, which indicates a constant overestimation of the random intercept variance in the results.

When examining the two-way interaction of the model with random slope reliability in Table 6 (using the convergence by model dataset), the bias means were lower when the reliability was 0.2, particularly in the within–between and within–contextual models (although these models lost about 47% of their original results due to convergence problems). When reliability was 0.6, the bias means in the within-only and random-conflated models were 7.94 and 7.65, respectively. Although these values could be interpreted as evidence of a large bias in the estimation of the random intercept variance in these models (whose true value was 4), we must consider that this parameter is not the same as in the DGM. Specifically, the variance of the random intercept in the DGM was conditioned on average units in average clusters (e.g., $x_{sc} - x_{.c} = 0$ at the cluster mean, and $x_{.c} = 0$, which was close to the grand mean). In contrast, this reference changes in the within-only and random-conflated models (i.e., the variance of the random intercept is conditioned solely on the within-cluster component or conflated components

of the level 1 predictor), making it difficult to know how much of the bias shown in Table 6 is actually bias and how much is a consequence of this change in the reference for the parameter.

Table 6. Bias Means for Random Intercept Variance in the Convergence by Model Dataset.

Factors	Bias mean	Proportion results
Random Slope Reliability = 0.2		
- Within-only Model	3.27	.98
- Within–between Model	0.44	.48
- Within–contextual Model	0.49	.47
- Random-conflated Model	2.48	.97
Random Slope Reliability = 0.6		
- Within-only Model	7.94	.99
- Within–between Model	1.62	.67
- Within–contextual Model	1.65	.68
- Random-conflated Model	7.65	.99
Random Slope Variance Predictor at Level 2[a]		
- 0.5	2.64	.79
- 1	4.05	.83
- 1.5	5.49	.85
Sample Size at Level 1		
- 5	6.18	.81
- 30	2.08	.84

Note. a = As described in Table 1, the true variance for the random slope of the predictor at level 2 was created as a proportion using the variance for the random slope of the level 1 predictor as a reference; The true value for the random intercept variance was 4.

Some of the differences found in the bias means of the random intercept variance across models may be related to the heterogeneity that the random slope reliability introduces into the

model, which is greater when 0.6 is used as a reference and would particularly affect models that do not disaggregate or include the random slope variance of the predictor at level 2. Regarding the levels of the random slope variance for the level 2 predictor in the convergence by model dataset, Table 6 shows that the bias mean for the random intercept variance increased as the size of intercept heteroscedasticity increased (which is consistent with the findings in the two-way interaction of the model with the reliability of the random slope). Moreover, the bias mean for the random intercept variance was about three times higher when level 1 sample size was 5 instead of 30.

In the convergence by replication dataset, the two-way interaction of model with sample size at level 1 showed a salient η^2 = 0.04, along with the two-way interaction of model with random slope reliability (η^2 = 0.05) and the random slope variance for the level 2 predictor (η^2 = 0.08). Table 7 shows these results. In general, the within-only and the random-conflated models registered the highest bias means across the levels of the factors (which is consistent with what we just described above), especially when sample size was small or slope reliability was high. Interestingly, the bias means of the within–between and within–contextual models with large sample size at level 1 were not only the smallest (around −0.10), but also the only negative means (indicating parameter underestimation). Regarding the levels of the random slope variance for the level 2 predictor in table 7, its bias means for the random intercept variance were very close to those shown in Table 6, even though around 50% of the replications were lost due to convergence problems.

Table 7. Bias Means for Random Intercept Variance in the Convergence by Replication Dataset.

Factors	Bias mean	Proportion results
Within-only Model		
- Sample Size at Level 1 = 5	10.12	.47
- Sample Size at Level 1 = 30	3.35	.53
- Random Slope Reliability = 0.2	3.66	.40
- Random Slope Reliability = 0.6	8.42	.61
Within–between Model		
- Sample Size at Level 1 = 5	2.61	.47
- Sample Size at Level 1 = 30	−0.11	.53
- Random Slope Reliability = 0.2	0.44	.39
- Random Slope Reliability = 0.6	1.66	.61
Within–contextual Model		
- Sample Size at Level 1 = 5	2.61	.47
- Sample Size at Level 1 = 30	−0.12	.53
- Random Slope Reliability = 0.2	0.44	.39
- Random Slope Reliability = 0.6	1.66	.61
Random-conflated Model		
- Sample Size at Level 1 = 5	7.78	.47
- Sample Size at Level 1 = 30	4.31	.53
- Random Slope Reliability = 0.2	2.69	.39
- Random Slope Reliability = 0.6	8.06	.61
Random Slope Variance Predictor at Level 2[a]		
- 0.5	2.57	.43
- 1	4.03	.52
- 1.5	5.45	.56

Note. a = As described in Table 1, the true variance for the random slope of the predictor at level 2 was created as a proportion using the variance for the random slope of the level 1 predictor as a reference; The true value for the random intercept variance was 4.

The next random effect is the random slope of the level 1 predictor, whose bias highlights the problems of fitting a model with conflated random slopes. As shown in Table 8, the random-conflated model showed the highest bias mean in the random slope variance of the level 1 predictor across datasets, whereas the correctly parameterized models showed negligible bias.

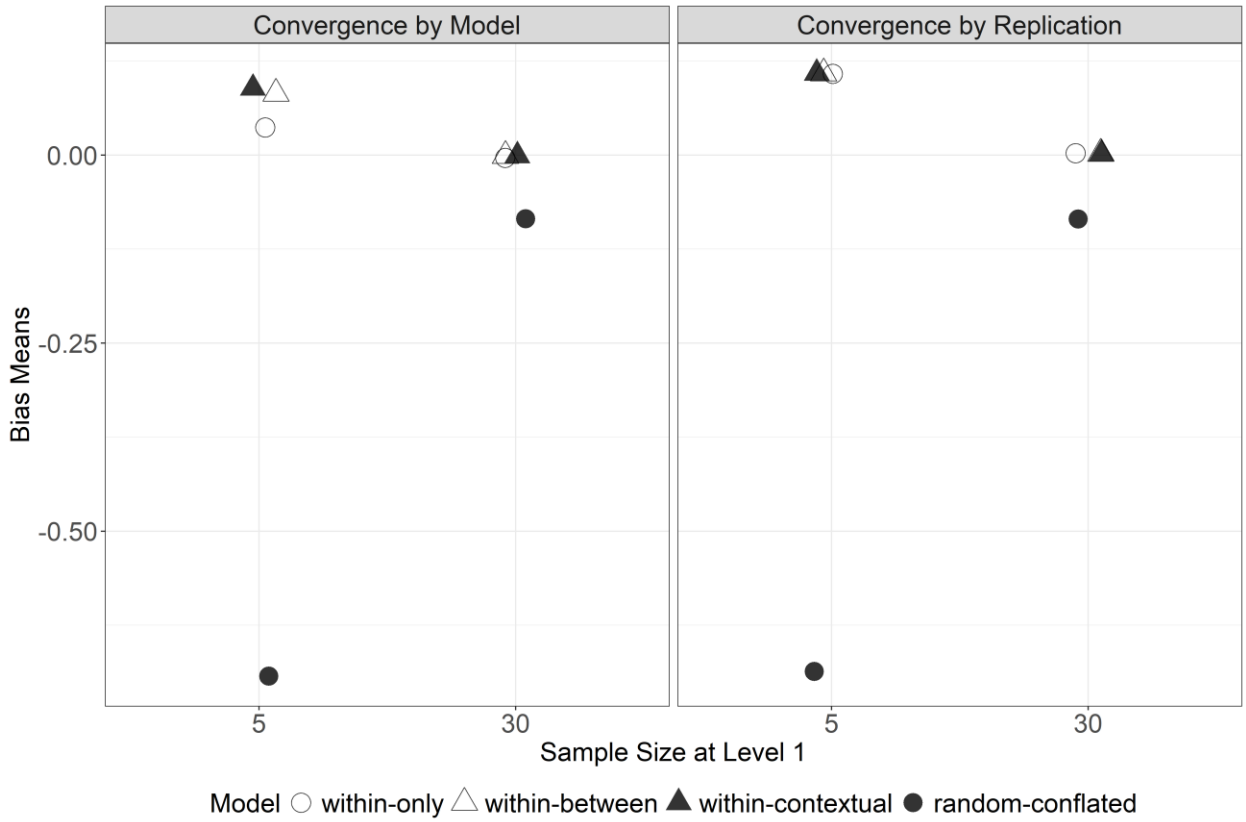
Table 8. Bias Means of the Random Slope Variance of the Predictor at Level 1.

Model	Convergence by Model	Convergence by Replication
- Within-only	0.02	0.05
- Within–between	0.04	0.05
- Within–contextual	0.04	0.05
- Random-conflated	−0.39	−0.37

Note. The true values for the random slope variance of the level 1 predictor were 0.5, 1.75, 2, and 4.5, which varied across the two levels of random slope reliability and the two levels of sample size at level 1, as shown in Table 1.

In the model for the bias of the random slope variance of the level 1 predictor, the two-way interaction of model with sample size at level 1 had the most salient η^2 across filtered datasets ($\eta^2 \approx 0.03$). Figure 4 shows these results. Even with the added jitter to avoid overlapping points, it is difficult to differentiate the bias of the correctly specified models, given that their accuracy in recovering the random slope variance was very similar, especially when the sample size at level 1 was 30 (with a bias mean < |0.10|). Although Figure 4 shows how the random slope bias increased in the random-conflated model for small level 1 samples sizes, it can also be seen how this problem becomes less noticeable as the sample size increased.

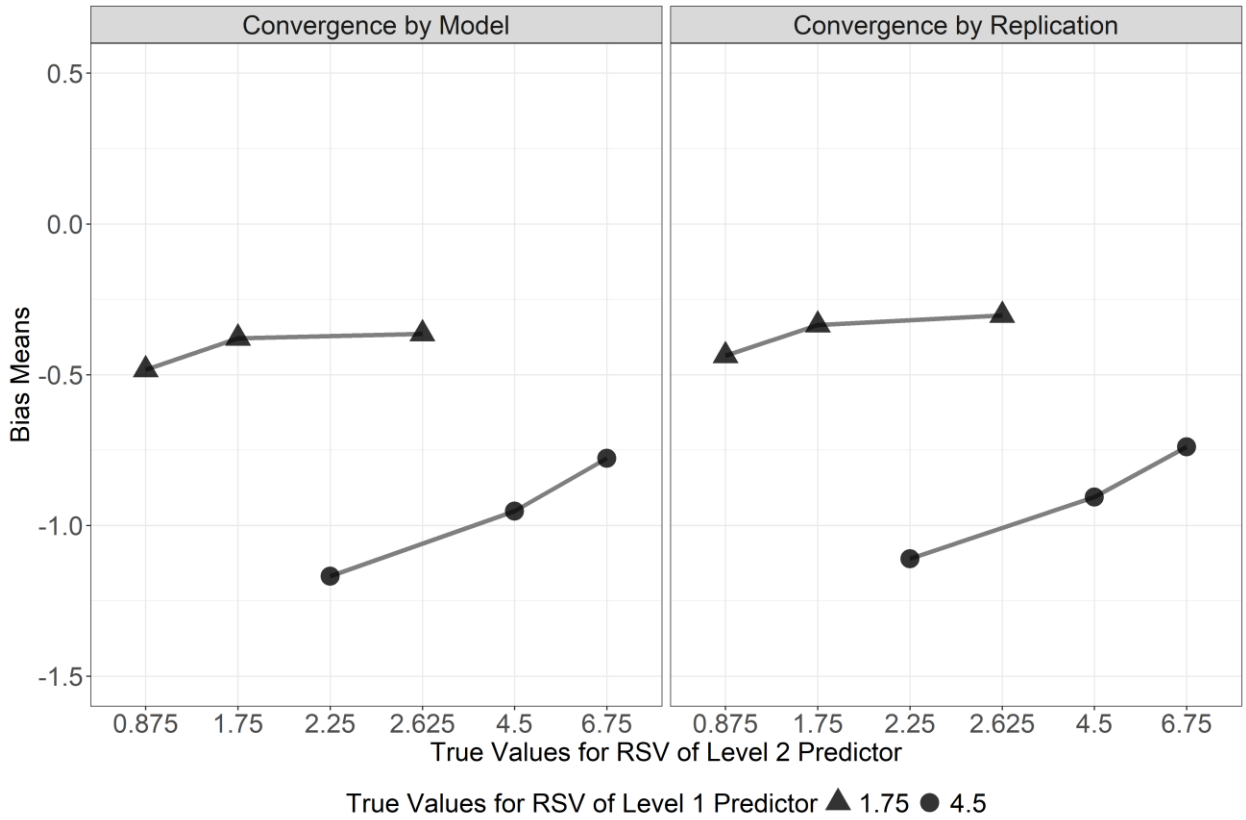
Figure 4. Bias Means of the Random Slope Variance of the Predictor at Level 1.



Note. Jitter was added to reduce overlap.

To further analyze the random-conflated model, we plotted its bias means according to the true values of the random slope variance of the level 1 and level 2 predictors, as the conflated random slope is a blend of these two sources of variance across clusters. Figure 5 shows these results. In general, the bias was larger when the true value for the random slope variance of the level 1 predictor was large (e.g., true variance = 4.5), especially when combined with small true values for the random slope variance of the level 2 predictor (e.g., around 2).

Figure 5. Bias Means of the Random Slope Variance for the Level 1 Predictor in the Random-conflated Model Across True Values for the Random Slope Variances.



Note. RSV = Random slope variance.

Results in Figure 5 are consistent with the problem of conflated random slopes: The bias in the random-conflated model is more noticeable as the variances diverge more, and the bias is negative because the variance of its conflated random slope is lower than it should be. Remember that the conflated random slope captures two sources of heterogeneity and its variance lies somewhere between the two, which is noticeable when the true variance for the random slope of the level 1 predictor was 4.5 (i.e., the black circles in figure 5) and 2.25 for the random slope of the level 2 predictor (x-axis in Figure 5), in which case the bias mean for the conflated random slope variance was approximately −1.2, which suggests that instead of recovering something close to 4.5, this parameter was close to 3.3 in the random-conflated model

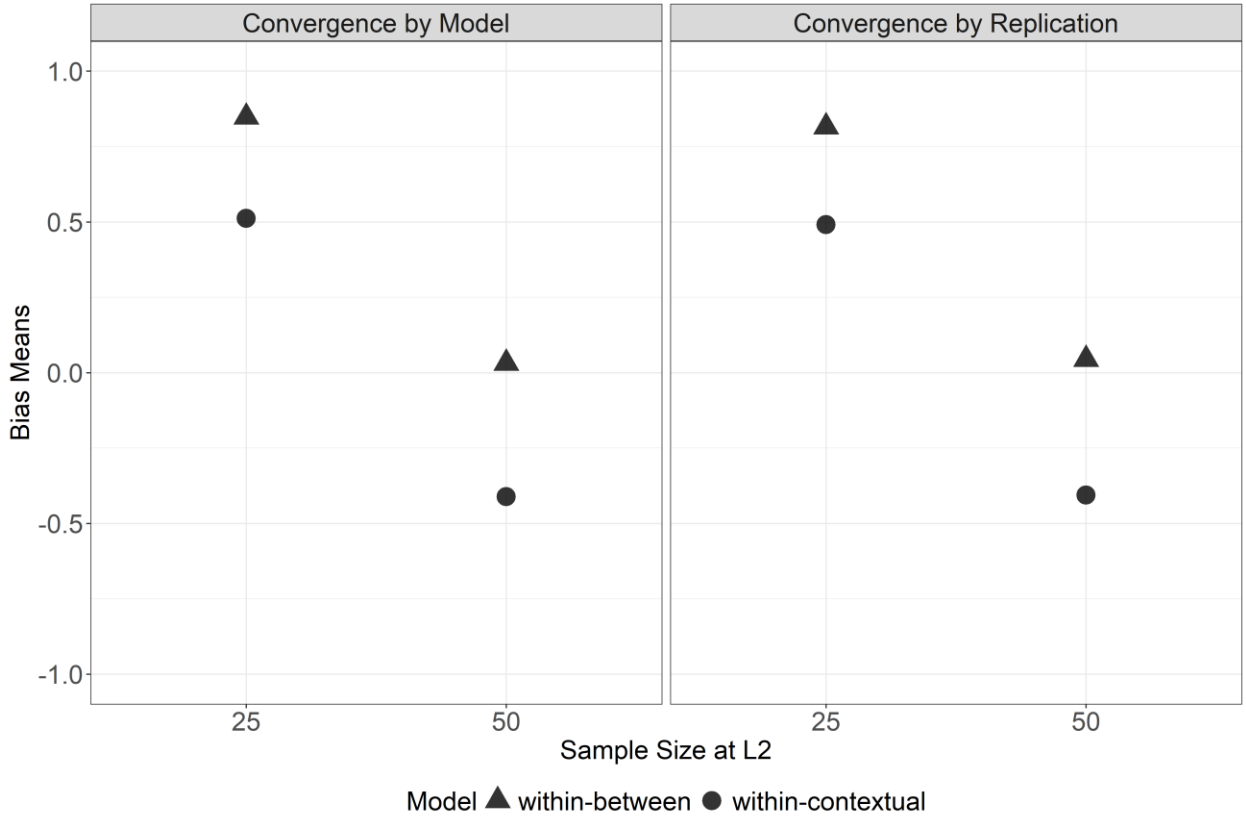
across datasets. In fact, the mean estimated variance of the conflated random slope was approximately 3.33 in the random-conflated model across datasets when the true variance for the random slope of the level 1 predictor was 4.5 and that for the level 2 predictors was 2.25.

The following results correspond to the bias means of the random slope for the level 2 predictor, which was available in the within–between and within–contextual models and whose function was to capture the heteroscedasticity of the cluster-intercepts across the values of the cluster-mean predictor at level 2. As the meaning of this parameter depends on the model (i.e., purely intercept heteroscedasticity in the within–between model vs differential or contextual intercept heteroscedasticity in the within–contextual model), we will compare its bias according to models.

In the within–between model, the sample size at level 2 had the most salient η^2 in the convergence by model dataset (e.g., η^2 = 0.03), whereas sample size at level 2 and the random slope reliability shown the most salient η^2 in the convergence by replication dataset (e.g., $\eta^2 \approx$ 0.02). In the within–contextual model, sample size at level 2 and random slope reliability showed the most salient η^2 in the convergence by model dataset ($\eta^2 \approx$ 0.03), whereas sample size at level 2 and the two-way interaction of sample size at level 1 with the random slope reliability showed the most salient η^2 in the convergence by replication dataset ($\eta^2 \approx$ 0.02).

Given that sample size at level 2 was relevant for the bias of the random slope variance of the level 2 predictor in the within–between and within–contextual models and across filtered datasets, bias mean differences according to this factor are depicted together in Figure 6.

Figure 6. Bias Means of the Random Slope Variance of the Predictor at Level 2.



The within–between model showed positive bias means in the estimation of the random slope variance of its level 2 predictor across sample size levels, and the within–contextual model shifted from positive to negative bias across the levels of the sample size at level 2. In both filtered datasets, the within–between model showed excellent performance recovering the random slope variance of the level 2 predictor when the sample size at level 2 was 50 (bias mean < 0.01), which means that, at least for this parameter and data condition, the model is working as it should, given that it is analytically equivalent to the model that generated the data.

Regarding random slope reliability in the model predicting bias in the random slope variance of the level 2 predictor using the convergence by replications dataset, bias means were 0.58 and 0.15 when reliability was set to 0.2 and 0.6, respectively. These results indicate that lower reliability—reflecting greater uncertainty in the estimation of cluster-specific slopes for

the level 1 predictor—is associated with greater upward bias in the estimated variance of the random slope of the level 2 predictor. Overall, the bias results for the within–between model across datasets suggest that insufficient precision in estimating random slopes (i.e., lack of precision in estimating the random slopes of the level 1 predictor across clusters), whether due to limited sample size or low reliability, can distort the estimation of variance components associated with level 2 predictors.

Table 9 shows the bias means for the random slope variance of the level 2 predictor in the within–contextual model according to the remaining prominent factors across filtered datasets. According to the convergence by model dataset (upper half of Table 9), the within–contextual model registered positive bias when the sample size at level 1 was 30 and when the random slope reliability was 0.2, whereas negative bias was found in the opposite scenarios.

Table 9. Bias Means of the Random Slope Variance of the Level 2 Predictor in the Within–Contextual Model.

Factors	Bias mean	Proportion results
Convergence by Model		
Random Slope Reliability		
- 0.2	0.44	.47
- 0.6	−0.40	.67
Sample Size at Level 1		
- 5	−0.62	.54
- 30	0.45	.61
Convergence by Replication		
Sample Size at Level 1 = 5		
- Random Slope Reliability = 0.2	0.24	.37
- Random Slope Reliability = 0.6	−1.26	.58

Sample Size at Level 1 = 30		
- Random Slope Reliability = 0.2	0.64	.41
- Random Slope Reliability = 0.6	0.30	.64

In the convergence by replication dataset (lower half of Table 9), differences in the bias means of the random slope variance of the level 2 predictor in the within–contextual model can be found according to the two-way interaction of sample size at level 1 and random slope reliability. These bias means dropped when a small sample size at level 1 was crossed with a low random slope reliability, whereas when the small sample size at level 1 was crossed with a high random slope reliability, the bias mean increased. These results may be linked to the effect of random slope reliability on the size of the true variance for the random slope of the level 2 predictor (i.e., the higher the reliability, the greater the variance), which in small samples at level 1 can cause distortion in parameter estimation (because there is not enough information to estimate large differences between clusters). This makes sense when looking at the bias means for sample sizes at level 1 equal to 30 in Table 9, given that in those cells the bias mean clearly decreases as the reliability of the slope increases, as the larger true variance and amount of information contribute to the parameter estimation.

Performance Measures: Power

The power of each model to recover the parameters of interest was analyzed using $-2\Delta\mathrm{LL}$ tests, where the change in log-likelihood of the converged models was evaluated against a mixture χ^2 distribution with degrees of freedom varying according to the number of parameters of the competing models. The statistical significance of these tests reflected the power of the models to recover the parameters of interest through the combinations of the design, then coded as a binary variable (0 = *p*-value ≥ .05, and 1 = *p*-value < .05). Furthermore, as the

focus of this study is on random slopes, only the power of the models to recover these parameters was analyzed. Table 10 shows the power of the models to recover the variances of the random slopes of the predictors at level 1 and level 2 (depending on which slope was available). In general, the four models had no problems recovering the random slope of the level 1 predictor (power ≈ .95). Regarding the random slope of the level 2 predictor, the within–between and within–contextual models achieved a power rate around .39 and .58, respectively, suggesting that within–contextual parameterization performed better than within–between in detecting this random slope. However, for these power rates we must consider that the true variance of the random slope of the level 2 predictor was always higher in the within–contextual model than in the within–between model, which may have contributed to the better performance of the former model over the latter. Additionally, it is worth mentioning that the power rates of the four models varied slightly across the filtered datasets, suggesting that the results are robust to the potential selection bias of listwise deletion.

Table 10. Power to Detect Random Slopes Across Models.

	Convergence by Model		Convergence by Replication	
Model	Power	Proportion results	Power	Proportion results
Random Slope of the Level 1 Predictor				
- Within-only	.94	.99	.97	.50
- Within–between	.97	.57	.97	.50
- Within–contextual	.96	.57	.97	.50
- Random-conflated	.94	.98	.97	.50
Random Slope of the Level 2 Predictor				
- Within–between	.39	.57	.40	.50
- Within–contextual	.58	.57	.60	.50

To analyze the effect of the factors on the power of the models to detect the random slope of the level 2 predictor, we fitted binary logistic regression models as described in Equation 12. Furthermore, due to the similarity across filtered datasets, we have only included the results based on the convergence by model dataset in the remainder of this subsection. Table 11 shows the most prominent factors for the power of the models with a random slope for the level 2 predictor.

Table 11. Prominent Factors for Detecting the Random Slope of the Level 2 Predictor Using the Convergence by Model Dataset.

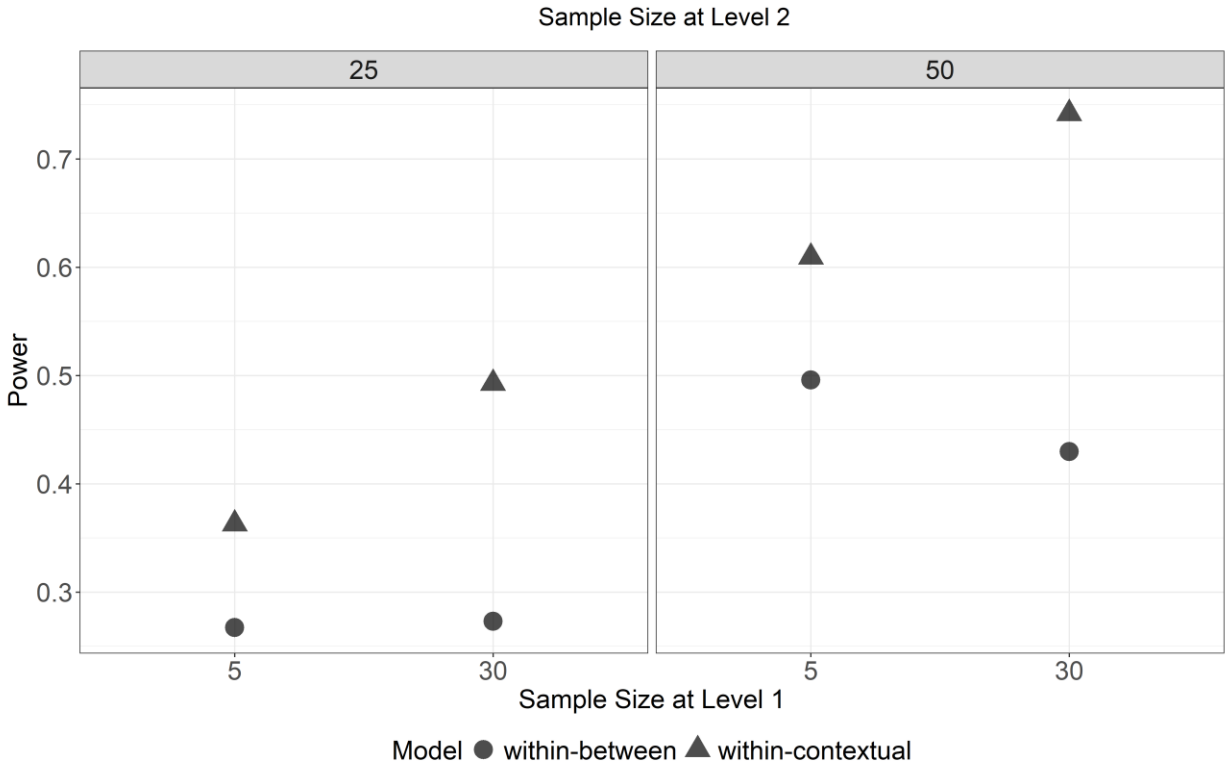
Factors	$-2\Delta LL$
Within–between Model	
- Sample Size at Level 1 x Sample Size at Level 2	**14.35**
- Sample Size at Level 2 x Random Slope Reliability	**4.43**
- Sample Size at Level 1 x Random Slope Reliability x Random Slope Variance Level 2 Predictor	**23.12**
Within–contextual Model	
- Sample Size at Level 1 x Sample Size at Level 2	**13.31**
- Sample Size at Level 2 x Random Slope Reliability x Random Slope Variance Level 2 Predictor	**8.07**

Note. Values in bold indicate *p*-values < .05.

The two-way interaction of the sample size at level 1 with sample size at level 2 was significant for the power of the within–between and within–contextual models. Figure 7 shows these results. The power of the within–between model to detect the random slope of its level 2 predictor was notably low in small samples at level 2, regardless of the sample size at level 1 (at both levels, the power was around .27). Although the power of this model increased when the size at level 2 was 50, its best performance only reached .49. As for the power of the within–

contextual model, this increased linearly across the sample size combinations, going from .40 in the most limited conditions to .72 in the best conditions.

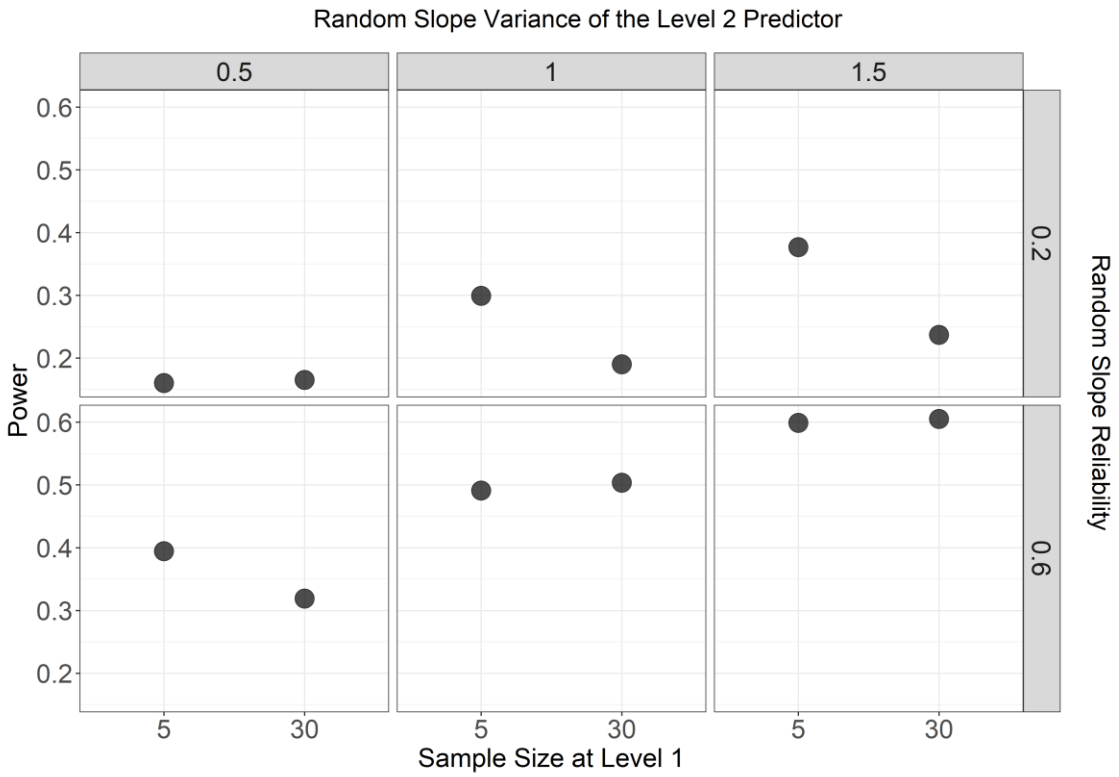
Figure 7. Power to Detect the Random Slope Variance of the Level 2 Predictor.



Regarding the more specific factors for the power of the within–between model, the three-way interaction of sample size at level 1 with the random slope reliability and the random slope variance for the level 2 predictor was significant, $-2\Delta LL(2) = 23.12$, $p < .01$. Figure 8 shows these results. The variation across level 1 samples was small in most combinations. As the variance of the random slope for the predictor at level 2 and the reliability of the random slope for the level 1 predictor increased, so did the power of the within–between model. Under the worst data conditions, its power to detect the random slope of the level 2 predictor was only .16, whereas under the best conditions it reached .59, with no differences with respect to sample sizes

at level 1. These results suggest that the power of the within–between model is related to the size of the intercept heteroscedasticity and the information about clusters present in the data.

Figure 8. Power to Detect the Random Slope Variance of the Level 2 Predictor in the Within–between Model.

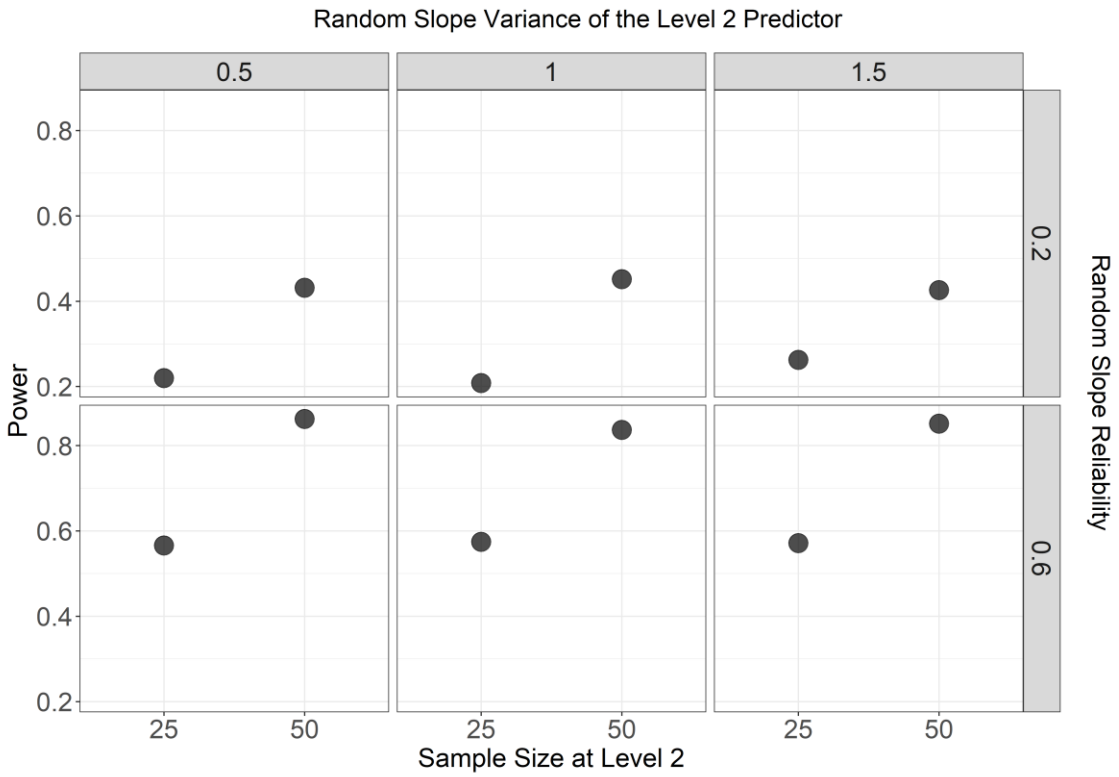


Finally, the differences in power in the within–contextual model according to the three-way interaction of sample size at level 2 with the random slope reliability and the random slope variance of the level 2 predictor, are shown in

Figure 9. There is a clear pattern in the cells of the figure: As the reliability of the random slope and the sample size at level 2 increased, so did the power of the model to detect the random slope of the level 2 predictor. This is consistent with some of the results described above and

with what was expected, given that the size of the random slope variance of a predictor and the samples at level 1 and level 2 should contribute to the estimation of parameters in the models.

Figure 9. Power to Detect the Random Slope Variance of the Level 2 Predictor in the Within–contextual Model.



Performance Measures: Empirical SE and SE Bias of the Fixed Effects

To conclude the Results section, the performance of the models was evaluated according to the empirical SE (as defined in Table 4) and SE bias of the fixed effects. Random effects were not analyzed in this subsection because the estimation of their standard deviations is based on a procedure different from that used for the SEs of fixed effects in the `lme4` package (Bates et al., 2015).

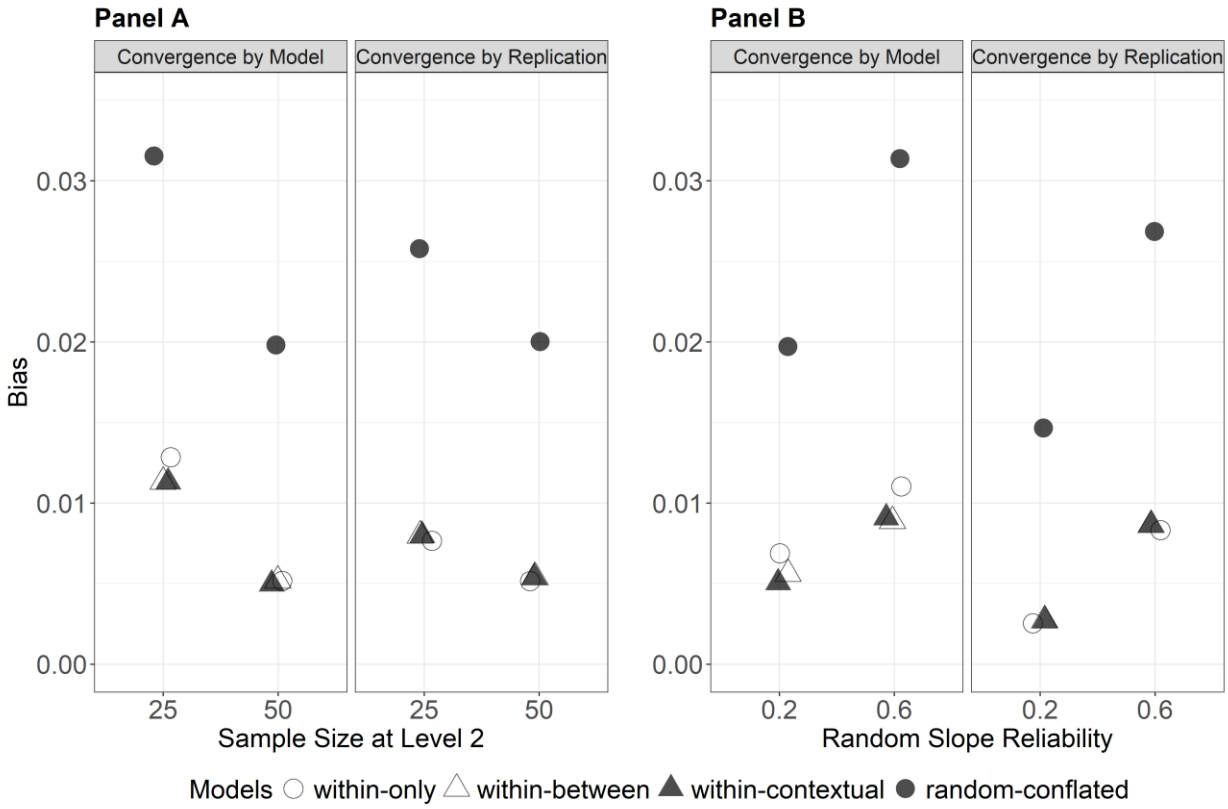
The empirical SE and SE bias were useful for examining the accuracy of models with respect to variability in the estimation of fixed effects across replications. Specifically, bias was analyzed using regression models identical to those described in Equation 13, in which the dependent variable SE bias was calculated by subtracting the model-specific SE of the estimated fixed effect in a given replication (i.e., what the variability of the fixed effect should have been according to the models) from the empirical SE of an estimated fixed effect across replications in a given combination of factor levels (i.e., reflecting what the actual sampling variability of the fixed effect was in the results).

The influence of design factors on SE bias for the fixed effects was evaluated using partial η^2 effect sizes. Positive values of SE bias indicate that the empirical SE—reflecting the actual variability of the fixed effect estimates across replications within a given condition—was greater than the model-based SE estimated within each replication. This suggests that the model tended to underestimate the true sampling variability. In contrast, negative SE bias values imply that the model-based SEs were larger than the empirical SE, indicating that the model overestimated the variability of the fixed effect relative to what was actually observed across replications. In both cases, the SE bias reflects the discrepancy between how much variation the models expect and how much variation was observed in the simulation results.

Regarding the within-cluster fixed effect, the salient factors in its bias model were the same across filtered datasets: the two-way interaction between model and random slope reliability ($\eta^2 \approx 0.04$), the two-way interaction between model and sample size at level 2 ($\eta^2 \approx 0.05$), and sample size at level 1 ($\eta^2 = 0.20$ and $\eta^2 \approx 0.16$ in the convergence by model and convergence by replication datasets, respectively). Figure 10 shows the bias means for the within-cluster fixed effect SE according to the two-way interactions.

In general, the bias means for the SE of the within-cluster fixed effect in Figure 10 show a small but consistent underestimation of the sampling variability of this parameter across models (i.e., the empirical SE was larger than the model-based SE). The bias across the filtered data sets in panels A and B in Figure 10 follows a similar pattern, although slightly lower bias means were found among the results filtered by replication—finding that could be the effect of listwise deletion that reduces the sampling distribution of the results. Panel A of Figure 10 shows how the bias means of the SE were higher among the models when the sample size at level 2 was 25, whereas Panel B shows how the bias increased among the models when the random slope reliability was 0.6—which can also be associated with the increase in the SE of the corresponding fixed effect that a random slope induces in the model.

Figure 10. Bias Means for the Within-cluster Fixed Effect SE.



Note. Jitter was added to reduce overlap.

Notably, the random-conflated model obtained the highest bias means in the SE of the within-cluster fixed effect in Figure 10, which is consistent with what we have discussed so far about conflated random slopes: The incorrect specification of the variance components in the model can affect the estimation of the SE of the fixed effects (the within-cluster fixed effect here).

For the between-cluster fixed effect, three factors showed a salient partial effect size in the convergence by model dataset: the two-way interaction of model with the random slope reliability (η^2 = 0.03), the sample size at level 2 (η^2 = 0.04), and the random slope variance of the level 2 predictor (η^2 = 0.06). These results are shown in Table 12. At first glance, the bias

means for the between-cluster fixed effect SE were smaller between models when the random slope reliability was 0.2 compared to 0.6. Interestingly, the within-only model registered the highest bias means in the SE of the between-cluster fixed effect across the two-way interaction cells, whereas the random-conflated model achieved a similar or lower bias than the other models across these cells.

Table 12. Bias Means for the Between-cluster Fixed Effect SE in the Convergence by Model Dataset.

Factors	Bias Means	Proportion results
Random Slope Reliability = 0.2		
- Within-only	0.08	.98
- Within–between	0.02	.48
- Within–contextual	0.02	.47
- Random-conflated	0.03	.97
Random Slope Reliability = 0.6		
- Within-only	0.15	.99
- Within–between	0.08	.67
- Within–contextual	0.08	.68
- Random-conflated	0.04	.99
Sample Size at Level 2		
- 25	0.10	.77
- 50	0.06	.88
Random Slope Variance Level 2 Predictor		
- 0.5	0.05	.79
- 1	0.08	.83
- 1.5	0.10	.85

Regarding the factors in the last two rows of Table 12, the SE bias of the between-cluster fixed effect was greater when the sample size at level 2 was 25, whereas as the reference for the

random slope variance of the level 2 predictor increased, so did the SE bias for the between-cluster fixed effect.

In the convergence by replication dataset, the model for the bias of the between-cluster fixed effect SE had similar salient factors to those analyzed in the convergence by model dataset, although without interactions. The pattern of results was similar to that shown in Table 12 (and thus, these results were not included in a new table): The bias mean of the between-cluster fixed effect SE was greater as the random slope variance of the level 2 predictor increased (up to η^2 = 0.10 for the highest level of this factor), when the random slope reliability was 0.6 (η^2 = 0.10), and in the within-only model (η^2 = 0.11).

Finally, we analyzed the bias of the contextual fixed effect SE, whose means in both filtered data sets were ≈ 0.09. In the convergence by model dataset, four factors showed salient effect sizes: random slope variance of the level 2 predictor (η^2 = 0.04), sample size at level 2 (η^2 = 0.04), random slope reliability (η^2 = 0.17), and model (η^2 = 0.19). These results are shown in Table 13. Interestingly, the bias means of the contextual fixed effect SE in the convergence by model dataset were similar to those shown in Table 12: As the random slope variance of the level 2 predictor increased, so did the bias. Furthermore, the bias of the contextual fixed effect SE increased when the sample size at level 2 was 25 instead of 50.

Table 13. Bias Means for the Contextual Fixed Effect SE in the Convergence by Model Dataset.

Factors	Bias means	Proportion results
Random Slope Variance Level 2 Predictor		
- 0.5	0.08	.79
- 1	0.10	.83
- 1.5	0.11	.85

Sample Size at Level 2		
- 25	0.12	.77
- 50	0.08	.88
Random Slope Reliability		
- 0.2	0.06	.78
- 0.6	0.13	.87
Model		
- Within-only	0.08	.99
- Within–between	0.05	.57
- Within–contextual	0.05	.58
- Random-conflated	0.13	.98

Regarding the random slope reliability in Table 13, it can be observed that when it was 0.6, the bias mean of the contextual fixed effect SE was approximately twice that found when the reference value for the reliability was 0.2. Moreover, the bias of this SE was higher in the random-conflated model, which may be linked to its incorrect specification of the random slopes (similar to what we found in the bias of the within-cluster fixed effect SE).

Regarding the bias of the contextual fixed effect SE in the convergence by replication dataset, three factors showed salient effect sizes: the random slope variance of the predictor at level 2 (η^2 = 0.03), the two-way interaction of the sample size at level 2 with reliability of the random slope (η^2 = 0.03), and the models (η^2 = 0.18). The bias means across the levels of the random slope variance for the predictor at level 2 and across the models were very similar to the results shown in Table 13 using the convergence by model dataset (e.g., differences in the third decimal place, so these results were not included in a new table), which may serve as evidence of the robustness of these results to the effect of listwise deletion.

The bias means of the contextual fixed effect SE according to the two-way interaction of sample size at level 2 and random slope reliability are shown in Table 14. The results suggest

that models underestimated the sampling variability of the contextual fixed effect, and that differences between level 2 sample sizes were not as notable as the differences according to levels of random slope reliability. While this may be linked to listwise deletion effects in the convergence by replication dataset, it may also suggest that for the bias of the contextual fixed effect SE, the precision in the estimation of random slopes is more relevant than the amount of sample available at level 2 (at least for levels 25 and 50 used in this study).

Table 14. Bias Means for the Contextual Fixed Effect SE in the Convergence by Replication Dataset.

Factors	Bias means	Proportion results
Random Slope Reliability = 0.2		
- Sample size at level 2 = 25	0.04	.27
- Sample size at level 2 = 50	0.05	.52
Random Slope Reliability = 0.6		
- Sample size at level 2 = 25	0.14	.45
- Sample size at level 2 = 50	0.10	.78

DISCUSSION

Key Findings

This study investigated the performance of four multilevel models in estimating unconflated random slopes under varying data conditions, focusing on bias, power, and SE accuracy. The results revealed distinct trade-offs across model specifications and provided strong evidence that conflated random slopes compromise the accuracy of both fixed and random effect estimation.

In terms of convergence, the within-only and random-conflated models exhibited high convergence rates (> .97), whereas the more complex within–between and within–contextual

models converged less frequently (in the .38–.73 range), particularly when both level 1 and level 2 sample sizes were small. These findings are consistent with previous literature (e.g., Rights & Sterba, 2023), which shows that models with fewer random components tend to converge more easily, though this often comes at the cost of correct model specifications or assumptions. For example, the within-only model has an unconflated random slope for the level 1 predictor, but it assumes homogeneity in the variance of cluster-specific intercepts, an assumption that can affect the accuracy of standard error estimates for fixed effects when the cluster-specific intercepts vary across the cluster-mean predictor values. The within–between and within–contextual models were much more likely to converge when both the level 1 and level 2 sample sizes were large, underscoring the importance of study design when fitting complex models.

Regarding estimation bias in the convergence by model and convergence by replication results, the within-only and random-conflated models consistently exhibited large bias in estimating the random intercept variance. However, these results should be analyzed with caution, given that the random intercept of these models is conditioned by variables other than those used to generate the data. This difference between the DGM model (with level-specific fixed and random effects) and models with only one random slope makes it difficult to interpret the bias results. For example, even though the within-only model showed bias means of around 8 in the convergence by model data for the random intercept variance when the reliability of the random slope was 0.6, when the random slope of the predictor at level 2 was included in the model, the bias decreased dramatically to around 1.6—although this reduction is not necessarily good news either, as this bias represents approximately 50% of the actual true value used for the random intercept variance.

As shown in Table 7, it is interesting to see how the bias in the estimation of the random intercept variance in the within-only model decreased when the sample at level 1 was large and there were small 'real' differences (i.e., a small random slope reliability) between the random slopes of the level 1 predictor across clusters. This emphasizes, on the one hand, the general recommendation to avoid small groups that can distort the estimation of random effects in the model (Snijders & Bosker, 2012) and, on the other hand, the effect that random slope heterogeneity can have on the inflation of the random intercept variance in the within-only model.

The within–between and within–contextual models provided more accurate fixed and random effects estimates, particularly under favorable data conditions (e.g., large samples at level 1 and level 2). Interestingly, when analyzing the random slope variance of the level 1 predictor, bias in the random-conflated model increased as the true variance of the level 1 slope increased and diverged from that of the level 2 slope, which illustrates the distortion that occurs when conflated random slopes are applied to data with distinct sources of slope heterogeneity. This is consistent with warnings from Rights et al. (2020) that conflating variance components can distort variance estimation, especially when those components differ in magnitude.

Regarding power, all models demonstrated adequate capability to detect the random slope of the level 1 predictor (≈ .95). The capability of the within–between and within–contextual models to detect the random slope of the level 2 predictor was moderate to low (≈ .40 and ≈.60, respectively). It is important to remember that although the results across the filtered datasets were consistent, for the within–between and within–contextual models we only had around 50–60% of the results, as the rest were discarded due to convergence problems. These convergence problems, linked to the criteria of the `lmer` package for evaluating the likelihood function and

the variance components of the model, pose another challenge for the problem of conflated random slopes related to the programming skills or knowledge of researchers in using other statistical packages or methods in the estimation of their models.

The within–between model's power to detect the random slope of the level 2 predictor was especially low in small level 2 samples (< .30), even when level 1 sample size was increased from 5 to 30. These results are aligned with Hamaker and Muthén (2020), who noted that identifying contextual or between-slope variation depends critically on having sufficient information about clusters. Our findings extend this idea by showing that model power is particularly sensitive to both the sample size at level 2 and the reliability of the random slope being estimated (although the analysis of the reliability of random slopes is not something that is frequently mentioned in the literature we have consulted).

Regarding SE bias, the models generally underestimated the empirical variability of the within-cluster fixed effect across replications, with the highest SE bias observed in the random-conflated model—especially under high random slope reliability and small sample sizes at level 2 (although these bias means were below 0.04). This indicates that the model underestimated uncertainty, consistent with McNeish's (2023) critique that oversimplified models may yield anti-conservative inferences (i.e., ignoring level 2 components inflates Type I error rates in the model). While all models showed some degree of SE underestimation, this bias was amplified in the random-conflated model, particularly for the within-cluster and contextual fixed effects, again reflecting misrepresentation of slope variance. Notably, SE bias for the between-cluster and contextual fixed effects increased with the reliability level of the random slope for the level 1 predictor across filtered datasets.

Taken together, these findings emphasize that models with conflated random slopes fail to recover critical variance components accurately, especially under conditions involving moderate random slope variance differences and moderate-to-high slope reliability. This study confirms the recommendations of recent literature (e.g., Rights & Sterba, 2023) that level-specific slope variances should be modeled explicitly whenever variance decomposition is of substantive interest. Our results further demonstrate that doing so is feasible and yields accurate parameter estimates and SEs when sample sizes of at least 30 units at level 1 and 50 clusters at level 2 are available.

Limitations and Future Directions

This study has some limitations that suggest directions for future research. First, the simulation focused exclusively on normally distributed continuous outcomes, limiting the generalizability of findings to non-normal, binary, or ordinal variables. Second, only frequentist models estimated using restricted maximum likelihood (REML) were evaluated; Bayesian approaches, which may yield more stable estimates in small samples or under complex model structures (Levy & Mislevy, 2016), could be explored. Third, while the simulation conditions were inspired by realistic scenarios, they did not incorporate irregular designs, such as severely unbalanced data or sparse cluster sizes, which are common in applied settings (Snijders & Bosker, 2012). Finally, the correlation between level 2 random effects in the DGM was fixed at .1 for simplicity; relaxing this constraint would allow for a more nuanced investigation of the equivalence and divergence of the within–between and within–contextual models.

Future work should expand both the scope of simulation conditions and the estimation methods. Incorporating categorical or skewed outcomes would offer insight into the behavior of conflated and unconflated random slopes under more complex data structures. Evaluating

Bayesian estimation could improve parameter recovery in small-sample or high-variance contexts where convergence is problematic for frequentist models (McNeish & Stapleton, 2016). Additionally, examining unequal cluster sizes and more extreme reliability conditions would help assess the robustness of slope variance recovery among the models examined.

CONCLUSION

The findings of this study highlight the importance of an issue that has been overlooked in modern multilevel analysis: Conflated random slopes can lead to substantial bias and distorted SEs, especially when slope heterogeneity and intercept heteroscedasticity are packed in raw level 1 predictors.

The random-conflated model showed high convergence rates and power for the random slope of the level 1 predictor but at the cost of misspecification. The within-only model also showed high levels of convergence and precision in the recovery of fixed effects and the variance of the random slope of the predictor at level 1, although it resulted in large SE bias of the fixed effects at level 2 (both the between-cluster and contextual fixed effects), probably due to the violation of its assumption of intercept homoscedasticity. The within–between and within–contextual models more accurately recovered the fixed and random components under adequate sample sizes, offering unbiased estimates and more trustworthy SEs when slope reliability was moderate to high. However, these models demanded larger level 2 samples to converge and to detect the random slope of the level 2 predictor reliably.

Importantly, this study suggests that modeling level-specific slope variances is both necessary and feasible when variance decomposition is a key inferential goal. The within–contextual model showed relatively better results than the within–between model in performance measures, so it may be a first choice for researchers. Because the variance components of both

models are analytically equivalent, what really matters is the purpose of the model (and the intended interpretation of its parameters). If the only thing that matters is the slope heterogeneity of the level 1 predictor, the within-only model is a good choice, although researchers should consider that the SE of the fixed effects at level 2 could have bias depending on the magnitude of the variability that is being left out of the model.

Finally, researchers are advised to explicitly mention the analysis of their level 1 predictors and the centering strategy (i.e., slope parametrization) used. Whenever possible, researchers should favor unconflated specifications and plan for sufficient level 1 and level 2 sample sizes to support their estimation.

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APPENDIX

Parallel of the Variance Components in the Within–between and Within–contextual Models

As Rights and Sterba (2023) explain, there is an analytical equivalence between the variance components of Models 2 (the within–between model in

Table 2) and 3 (the within–contextual model in

Table 2), in which the variance of the random slopes of each can be obtained by reparameterizing the other.

Using the within–between model as a reference, we have the following variance–covariance matrix for the random effects at level 2:

$$\begin{bmatrix} U_{0c} \\ U_{2c} \\ U_{3c}^{b} \end{bmatrix} \sim MVN\left(\begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} \tau_{U_0}^2 & \tau_{02} & \tau_{03}^{b} \\ \tau_{02} & \tau_{U_2}^2 & \tau_{23}^{b} \\ \tau_{03}^{b} & \tau_{23}^{b} & \tau_{U_3^b}^2 \end{bmatrix} \right)$$

It should be noted that this model uses $x_{sc} - x_{.c}$ at level 1 and $x_{.c}$ at level 2 to estimate random slopes, which creates level-specific effects, so each random slope can be interpreted as purely slope heterogeneity (U_{2c}) and purely intercept heteroscedasticity (U_{3c}^{b}) for a given cluster c.

The within–contextual model has the following variance–covariance matrix for its random effects at level 2:

$$\begin{bmatrix} U_{0c} \\ U_{1c} \\ U_{3c}^{c} \end{bmatrix} \sim MVN\left(\begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} \tau_{U_0}^2 & \tau_{01} & \tau_{03}^{c} \\ \tau_{01} & \tau_{U_1}^2 & \tau_{13}^{c} \\ \tau_{03}^{c} & \tau_{13}^{c} & \tau_{U_3^c}^2 \end{bmatrix} \right)$$

In this model, x_{sc} at level 1 and $x_{.c}$ at level 2 are used to estimate random slopes, which creates mutually controlling effects. The random slope U_{1c} captures the heterogeneity of the within-cluster fixed effect of x_{sc} across clusters, and U_{3c}^{c} captures the differential heteroscedasticity of the intercept for a given cluster c.

Although our notation may suggest that only the variance of the random intercept is the same across the within–between and within–contextual models ($\tau_{U_0}^2$), both the variance of the random slope for the level 1 predictor and the covariance between this random slope and the random intercept are the same between models (i.e., $\tau_{U_2}^2 = \tau_{U_1}^2$ and $\tau_{02} = \tau_{01}$). The analytical equivalence between the remaining components of the variance works as follows:

Variance components from the within–contextual model based on the results of the within–between model:

$$\tau_{03}^{c} = \tau_{03}^{b} - \tau_{02}$$

$$\tau_{13}^{c} = \tau_{23}^{b} - \tau_{U_2}^2$$

$$\tau_{U_3^c}^2 = \tau_{U_3^b}^2 + \tau_{U_2}^2 - 2(\tau_{23}^{b})$$

Variance components from the within–between model based on the results of the within–contextual model:

$$\tau_{03}^{b} = \tau_{01} + \tau_{03}^{c}$$

$$\tau_{23}^{b} = \tau_{U_1}^2 + \tau_{U_3^c}^2$$

$$\tau_{U_3^b}^2 = \tau_{U_1}^2 + \tau_{U_3^c}^2 + 2(\tau_{13}^{c})$$

These linear combinations arise from the general formula for the variance of a sum of random variables:

$$\mathrm{Var}(X+Y) = \mathrm{Var}(X) + \mathrm{Var}(Y) + 2 \times \mathrm{Cov}(X,Y)$$

and its rearranged form:

$$\mathrm{Var}(X-Y) = \mathrm{Var}(X) + \mathrm{Var}(Y) - 2 \times \mathrm{Cov}(X,Y)$$

For example, although in the within–contextual model it may seem counterintuitive to combine variances from the within–between model, this is justified by the algebra of random effects. In other words, $\tau^2_{U_3^c}$ is mathematically equal to the difference of $\tau^2_{U_3^b}$ and $\tau^2_{U_2}$, which implies:

$$\tau^2_{U_3^c} = \tau^2_{U_3^b} - \tau^2_{U_2} = \tau^2_{U_3^b} + \tau^2_{U_2} - 2(\tau^b_{23})$$

Finally, as Rights and Sterba (2023) suggest, the choice of model depends, among other factors, on the interpretation sought by the researcher for each parameter, as both models estimate unconflated random slopes.

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